

When Silver Isn't Gold: Quantifying and Auditing Group-Based Disagreement in Human and LLM Annotation

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LLM-as-judge systems are increasingly used as scalable substitutes for human annotators, including in polarized, subjective tasks where human annotators systematically disagree along group lines. Whether these systems reproduce, ignore, or distort human polarization patterns is largely untested, in part because existing methods for attributing this disagreement may be unreliable in realistic and subjective annotation settings. In this work, we find that their unreliability partially rests on the faulty assumptions that all systematic disagreement can be attributed to at least one annotator group and that each such group has a uniform bias across items. Instead of using these methodologies, we develop a method following a clustering approach that resolves both failure modes, enabling this to be quantifiably attributed to any annotator group. Using this method, we audit LLM-as-judge systems against real human annotation datasets and find that their disagreement patterns are largely disconnected from those of human annotators, a gap that persists even when models are explicitly prompted to condition on persona characteristics. We validate the underlying method on simulated data, showing it remains reliable under group-size imbalance and disagreement with conflicting directions that degrade current approaches. Overall, our results suggest that persona-conditioning alone is insufficient for LLM judges to replicate the structure of human disagreement.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; **Unsupervised learning**; Language resources; • **Human-centered computing** → **Empirical studies in collaborative and social computing**; • **Social and professional topics** → *Race and ethnicity*; *Gender*; *Sexual orientation*; *Age*.

Additional Key Words and Phrases: Polarization, Clustering, Annotation, Sociodemographic Characteristics, Persona-Prompting

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1 Introduction

As human annotation is generally expensive, LLM-as-judge systems are increasingly deployed as scalable substitutes for human annotators [23, 39] to acquire a cheaper, “silver” supervision signal in place of human-annotated “gold” labels. Recent work [14, 33, 35] shows that ignoring disagreement in human annotations may provide us with the wrong signal, which can lead to significant misconfiguration in trained AI systems [45]. For example, toxicity models have disproportionately marked African American (AA) speech as “toxic” [41], or systematically ignored minority opinions [37] in the past. It is therefore imperative that systematic disagreements by human annotators be preserved by LLMs whose annotations by default gravitate toward the statistical average [5]. Persona-prompting is sometimes used to bypass this limitation, but suffers from its own challenges such as stereotyping and inconsistent annotations [28, 30]. However, whether current LLM-as-judge systems reproduce, ignore, or distort these human disagreement patterns

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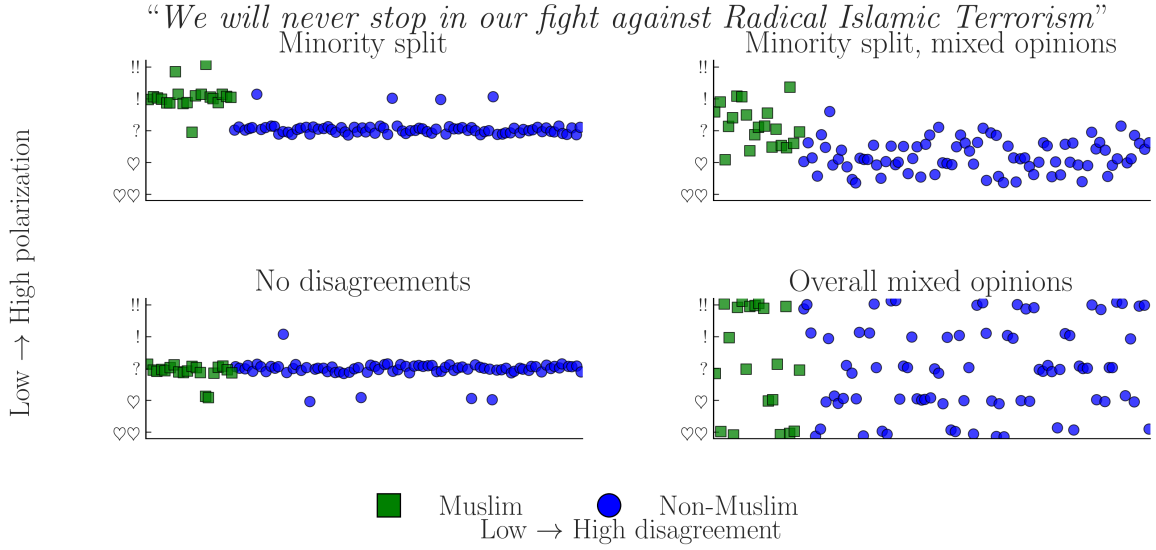


Fig. 1. Simulated experiment exploring four scenarios of a HS detection task applied to a controversial statement. The variance in annotations across different annotators may cause traditional IAA metrics to ignore minority opinions, especially when opinions are mixed (top-right).

is largely untested. We show that this is in part because the tools available for measuring group-based disagreement attribution are themselves unreliable.

In subjective tasks such as content moderation, toxicity, and Hate Speech (HS) detection, many approaches attempt to distinguish between general and systematic disagreement (Fig. 1), as research has shown that the latter may be caused by different points-of-view between groups [16, 33, 45]. This distinction typically relies on Inter-Annotator Agreement (IAA) [22, 24, 46], although other methods have been proposed [7, 35]. These systematic, fundamental disagreements are related to the phenomenon of *polarization*, where opinions are split between two conflicting groups.

Prior work has framed the question of LLM annotation reliability in terms of human-LLM agreement [10, 15, 31] or task-specific performance [20]. Instead, we posit that in order for the silver LLM signal to be representative of the gold human signal, the fundamental, systematic disagreements in the annotations should be similar. Thus, we ask the following **Main Research Question (MRQ): To what extent do LLMs follow human disagreement patterns in subjective annotation tasks?** We attempt to answer this question irrespective of whether this disagreement corresponds to the annotators’ actual group identities; we instead explore it through the following sub-questions:

RQ 1: When and why do current methods for systematic disagreement attribution fail? We find that prior approaches generally fail in realistic annotation scenarios (Finding 1). Statistical methods fail as systematic disagreement across items has a *direction*, preventing aggregation (Finding 2), while another clustering-based method wrongly assumes that all disagreement can be explained and attributed to some annotator characteristic, which we find not to be the case through the exhaustive creation of theoretical annotator groups (Finding 3).

RQ 2: How can we quantitatively attribute systematic disagreement to annotator groups? We introduce a new metric “*apunim*”, which *quantitatively* attributes systematic disagreement to specific Personal Characteristics (PCs), outperforming prior approaches in realistic annotation scenarios (Fig. 2). By applying it to four subjective Natural Language

Processing (NLP) datasets we confirm that annotator gender and ethnicity are sources of systematic disagreement (Finding 4).

RQ 3: *Which systematic disagreement patterns are shared between human and LLM annotators?* We conduct an annotation study on one Toxicity and one HS dataset using six open-source models in order to compare systematic disagreement patterns between LLM groups with equivalent disagreement exhibited between human groups (§6). We find that models using persona-prompting generally fail to reproduce any systematic subgroup disagreement patterns regardless of instruction prompt, confirming literature suggesting that persona prompting does not consistently influence LLM outputs in the examined tasks (Finding 5).

Our work makes the following contributions: (1) The identification and diagnosis of some previously unknown severe issues present in prior disagreement attribution methodologies. (2) A new attribution metric that disambiguates systematic from general disagreement and is robust to small annotator counts, realistic subgroup disagreement effects and imbalanced group splits which are common when analyzing minority annotations. (3) An evaluation on LLM disagreement patterns attributable to PCs included in their persona prompts in subjective annotation scenarios. We also release a PyPI library implementing our metric to aid future research in this topic.

Content Warning: Controversial and potentially harmful speech used for demonstration.

2 Background and Related Work

Terminology. We distinguish three related definitions used throughout this work. (1) **Disagreement** represents any variance among annotators' ratings for the same item, regardless of its source. (2) **Systematic disagreement** is the subset of disagreement that is non-random and recurring, which in this work is assumed to be caused by differences in group identity (e.g., background, cultural context, sociodemographic characteristics). (3) **Polarization** is the underlying societal phenomenon describing genuinely conflicting, opposed viewpoints on the same item.

Agreement. Current work on analyzing subjective tasks rests on using IAA between groups of annotators [27, 33, 44]. IAA metrics such as Cohen's kappa, Krippendorff's alpha, and Fleiss's kappa aim to distinguish genuine disagreement from random disagreement through statistical chance-correction. However, these metrics are often difficult to apply to intergroup IAA without extensions or restrictive assumptions (e.g., no missing labels or equal numbers of annotations). They have also been criticized for failing to account for differences arising from social, cultural, and demographic backgrounds [6, 7, 13, 46], producing results that are difficult to compare across studies, being sensitive to class imbalance [13], and relying on questionable assumptions about chance agreement [7]. While recent work has addressed some of these limitations with methods that improve applicability and robustness [22, 46, 51], they are much more complex and less intuitive compared to their traditional counterparts, indicating that IAA may not be the correct tool for analysis of minority viewpoints. Ultimately, IAA analysis may be useful for exploration and identification of some controversial items, but fails to explain the reason why they are controversial [1, 47].

LLM Annotation. Prior work suggests that LLMs are not necessarily reliable annotators when prompted with personas, as they may reproduce stereotypes [28, 30] and perform unreliably for minority groups [30]. To address subjectivity in annotation, several NLP approaches incorporate annotator metadata into LLM fine-tuning [9, 14, 19, 40], although evidence suggests that such improvements may be specific to the datasets on which models are trained [26]. An alternative line of work recovers annotator perspectives through clustering in embedding space [48], although these representations can be difficult to interpret, computationally expensive to obtain and evaluate, and sensitive to the

quality of the underlying embeddings. We instead posit that directly clustering the annotations provides a comparatively simple and interpretable mechanism.

Polarization. There are multiple competing formulations for estimating polarization besides IAA. One of these approaches [2] utilizes χ^2 to estimate in-group vs out-group variance. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric (GMM) [7], or non-parametric [32, 35] approaches to the annotation histograms. For our work, we use a non-parametric disagreement metric, normalized Distance From Unimodality (nDFU), which has been shown to correlate better with human judgment of disagreement than established metrics [35], although we compare prior approaches in Fig. 2. nDFU (Eq. 1) takes values in the range [0, 1], where values close to 0 indicate no polarization (i.e., opinions are concentrated around a single mode). Intuitively, nDFU measures how prominent secondary peaks are compared to the main peak (see Fig. 4, left for an example). The A Posteriori Unimodality (AU) method [35] builds upon nDFU stating that if a single item’s annotations are multimodal, but unimodal for all annotator subgroups, then polarization is explained by those groups. Though promising, this mechanism operates on the assumption that *all* polarization can be explained by a single split, which we prove to be impossible in many cases (§4).

$$\text{nDFU}(X) = \max_{i \neq m} \left(f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right) / f_m, m = \arg \max_i f_i \quad (1)$$

where, X denotes the set of annotations, f_i the relative frequency of the i -th annotation (e.g., the frequency of 2 in a scale of 1–5), and m the histogram peak.

3 Datasets

Table 1. Datasets used in this study.

Dataset	#Comments	#Annotators / Item	#PCs	Task
Kumar [25]	1000	5	10	Toxicity
Sap [42]	585	5–6	3	Hate Speech
DICES-350 [3]	350	123	4	Hate Speech
DICES-990 [3]	990	70–76	4	Hate Speech
POPQUORN [36]	1500	7–9	5	Hate Speech

We use five datasets from current NLP research that contain PC information at the level of individual annotators.

- The *Kumar* dataset [25] which contains annotations for Toxicity detection.
- The *Sap* “Breadth of Posts” dataset with the “*toyou*” labels [42] for HS detection.
- The *DICES-350* (*Q3_bias_targeting_inherited_attributes* label) and *DICES-990* (*Q3_bias_incites_hatred* label) datasets [3], which pertain to LLM response safety. Our choice of labels allows us to reframe these datasets as HS datasets, capturing conceptually similar dimensions despite the lack of shared label names across the datasets.
- The POPQUORN [36] dataset using the “offensiveness” label for HS detection.¹

¹While the dataset also contains “politeness” labels, these may not directly translate to toxicity detection; comments can simultaneously be toxic and phrased politely. Indeed, toxicity is not mentioned in the methodology of the original POPQUORN dataset and is considered a separate NLP task in other work [29, 53].

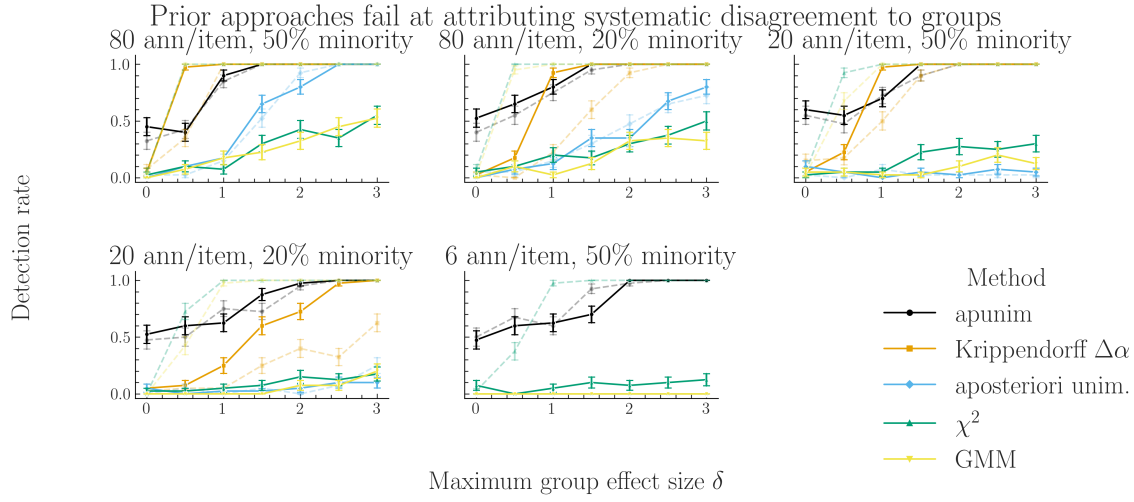


Fig. 2. Polarization group attribution on synthetic distributions. Bars indicate standard error. We assign a number of annotators per item, and split the annotator groups 50-50 or 80-20. Higher δ represents higher polarization. Apunim is adept at attributing systematic disagreements to the correct group regardless of number of annotators per item, or how these annotators are split. Dashed lines show the results of a simplified experimental setup, where disagreement always has the same direction.

A brief overview of the dataset characteristics is provided in Table 1. The Kumar dataset is notable for its large scale, featuring over 100,000 annotated items and 10 distinct PC dimensions. In contrast, the DICES datasets have fewer annotated items but involve more than an order of magnitude more annotators per item than the others. The POPQUORN dataset presents a modest number of annotators per item (7-9) and

Further details on the annotations, including overall levels of observed polarization (nDFU), are available in App. B and a discussion on the importance of high annotator counts per item, as opposed to generally high annotator counts can be found in App. J.

The full datasets are used for the human annotation analysis in §5 and their PC labels and texts as input for the LLM annotation in §6. In order to allow for a full comparison between human and LLM annotations, we use a sample of 1,000 items from the Kumar dataset. Our results remain consistent for both different samples and differently-sized samples (App. H).

4 Why Do Current Methods Fail?

4.1 Method Comparison

We compare the effectiveness of our method (introduced in §5), the AU framework proposed by Pavlopoulos and Likas [35], the χ^2 method presented in Akhtar et al. [2], GMM clustering used in Checco et al. [7], and a baseline using intergroup disagreement ($\Delta\alpha$ Krippendorff) on simulated distributions. To determine whether a score is statistically significant, we construct a null distribution for each method with the null hypothesis that the group labels are unrelated to the observed annotations. We therefore repeatedly shuffle the group labels across annotators, recompute the method's statistic, and use the resulting scores to form the null distribution. The observed score is then compared with this distribution.

Annotations are generated according to $y_{ij} = b_j + g_i p_j + \epsilon_{ij}$, where b_j is the baseline rating for item j , $g_i \in \{-\frac{1}{2}, +\frac{1}{2}\}$ indicates the annotator's group, p_j controls the magnitude and direction of systematic disagreement for item j , and ϵ_{ij} is annotation noise. We try two simulation setups that differ only in how p_j is defined: (1) **Unidirectional**: $p_j \sim \text{Uniform}(0, \delta)$. The group effect always favours the same group across all items, though its magnitude varies per item. (2) **Bidirectional**: $p_j \sim \text{Uniform}(-\delta, +\delta)$. The direction of the group effect can vary across items, so that groups may systematically disagree in opposite directions on different items (see Finding 2).

As shown in Fig. 2, both the χ^2 and GMM methods generally fail at attributing systematic disagreements to the correct group. IAA remains competitive when many annotators are available, but underperforms in cases where the examined group is a minority and when a few annotators are available per item (a common case among NLP datasets—see §3).²

FINDING 1. *Proposed methodologies attributing systematic disagreements to groups generally fail in realistic annotation conditions.*

4.2 Conflicting Disagreement Directions

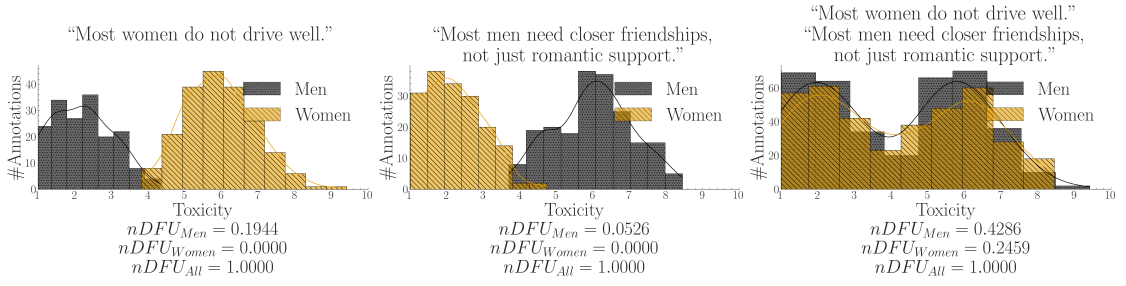


Fig. 3. Synthetic experiment showing two comments with annotators disagreeing about which one is toxic. Instead of comparing two unimodal distributions (comments viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions during aggregation, obscuring the source of the polarization (explained by gender, but not shown).

In Fig. 2 we observed that the simple statistical approaches do indeed show excellent performance when all groups are polarized in the same direction. This is expected, since they rely on consistent fixed means effects, which is disrupted by opposing disagreement effects violating their assumptions on fixed mean differences between groups. Specifically, both χ^2 and GMM aggregate across items before looking for group differences; χ^2 pools all annotations into a single contingency table, while GMM clusters annotators by their mean rating across items. In both cases, opposite-direction effects cancel out in the aggregate, making the two groups appear statistically indistinguishable. Thus, while these methods perform well in ideal conditions, they may completely fail on realistic annotation datasets.

We can confirm the persistence of this issue using nDFU [35], which is another representation of polarization (see §2). Fig. 3 shows that two items polarized for the opposite reason may lead to a false negative. We avoid this issue in §5 by operating item-by-item, asking whether the annotation distribution becomes unimodal when conditioned on group membership, without assuming a consistent direction of disagreement across items.

FINDING 2. *Systematic disagreements may conflict across items, which may compromise cross-item annotation aggregation.*

²Ablations with $k = \{3, 5, 10\}$ groups can be found in App. H; Fig. 7.

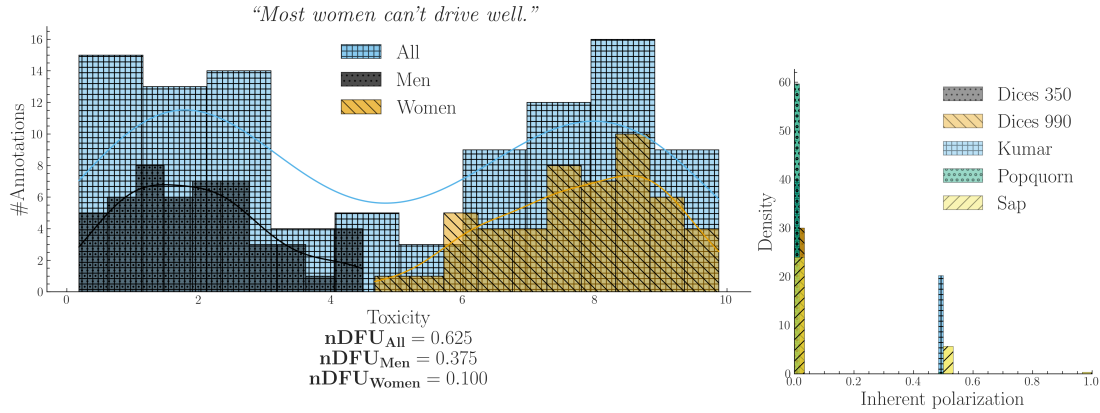


Fig. 4. **Left:** Simulated experiment describing a polarizing comment where men and women agree between themselves, but disagree with the opposite gender. **Right:** Unattributable Polarization (UnPol) across our datasets (§3). Contrary to assumptions made in prior work, UnPol is commonly non-zero and can even reach $nDFU = 1$ in datasets with low annotator counts.

4.3 Inherent Polarization

Opposing disagreement may explain the failure of statistical methods, but not the proposed AU method which operates only on a single item's annotation histogram. The under-performance of AU could instead be attributed to a violation of its assumptions. Indeed, a newly discovered phenomenon we call UnPol comes in direct contrast to the assumption that all systematic disagreement can be explained by a split in annotations.³

What Is It? UnPol refers to disagreement that cannot be attributed to *any arbitrary* group. AU attributes polarization to groups only when all each individual groups show completely unimodal annotation histograms. For example, in Fig. 4 left, AU would only attribute polarization if $nDFU(all) > nDFU(men) = nDFU(women) = 0$. The presence of UnPol would invalidate this assumption, since there would be no possible group or combination of groups g s.t. $nDFU(g) = 0$.⁴

Formally, the UnPol for a single item c is the minimum $nDFU$ exhibited by any arbitrary group (Eq. 2). When considered exhaustively, these theoretical, random splits produce both observed and latent groups. Since $nDFU$ operates only when a group has at least three annotators (Eq. 1), this yields $\sum_{k=3}^n \binom{n}{k}$ possible groups, which is computationally intractable for large annotator counts. We can therefore use Monte Carlo sampling by drawing k random partitions with random sizes (Eq. 3) which serves as an upper bound (as it considers a strict subset of Eq. 2).

$$\text{Pol}_{inh}(c) = \min_{S \in \mathcal{G}_c} \{nDFU(S)\} \quad (2) \quad \text{Pol}_{inh}^{MC}(c) = \min_{j=1}^k \{nDFU(S_j)\} \quad (3)$$

where $\mathcal{G}_c$ is the collection of all subsets S of annotations for an item c such that $|S| \geq 3$ and S_j is a random sample drawn from $\mathcal{G}_c$ for $j = 1, \dots, k$.

Does It Exist? We compute the UnPol for each item across the four datasets. For Sap and Kumar, we use the exhaustive formulation (Eq. 2), whereas for the two DICES datasets we rely on the Monte Carlo approximation with $k = 1000$ (Eq. 3). Fig. 4 (right) shows that in all DICES items there exists at least one group that fully accounts for the observed

³Given the terminology in §2 UnPol is an inaccurate term as it does not pertain to

⁴Fig. 4 also shows a clear failure mode of AU regardless of assumptions. While it is obvious that in the example polarization is caused by gender, AU expects total unimodality to attribute it.

nDFU even when repeatedly subsampling down to 6 annotators.⁵⁶ Interestingly, this observation does not hold for the Kumar and Sap datasets, despite evaluating all possible annotator groups exhaustively. Therefore, even if UnPol was not responsible for the degraded performance of AU in our synthetic data, it would certainly have an impact when applied in real-world datasets. See App. B; Table 9 for examples of items with high UnPol.

FINDING 3. *Items may exhibit a degree of UnPol that cannot be attributed to any known or latent annotator grouping.*

5 Quantifying Attribution to Annotator Groups

5.1 Our Methodology

Instead of using UnPol, which represents the minimum possible systematic disagreement exhibited by any group, we calculate the *expected* (average). Therefore, we do not test whether the group is polarized against the whole; nor whether the group explains *some polarization* (as would be the case if we used Eq. 2), but rather whether the group explains polarization *more than other groups*.

Specifically, we define apriori polarization for an item c as $\text{Pol}_{apr}(c) = \mathbb{E}_{S \sim \mathcal{S}_c} [\text{nDFU}(S)]$, which is averaged across items for a dataset $\mathcal{D}$. Intuitively, a high $\text{Pol}_{apr}(c)$ score means that much of the polarization is likely not driven by any single group. We compare this quantity with the average observed nDFU of all items, which yields our normalized metric, *apunim* (Eq. 4). We explain why defining $\text{Pol}_{apr}(c)$ in this way is preferable to alternatives in App. E. The *sign* of *apunim* carries information agreement-based attribution does not:

- $\text{apunim}(g) \approx 0$: The group g does not meaningfully disagree with the other annotators.
- $\text{apunim}(g) > 0$: The group g has fundamental disagreements with the other annotators.
- $\text{apunim}(g) < 0$: The annotators of group g are polarized between themselves.

$$\text{apunim}(\mathcal{D}, g) = \frac{\mathbb{E}_{c \in \mathcal{D}} [\text{Pol}_{apr}(c)] - \mathbb{E}_{c \in \mathcal{D}} [\text{Pol}_g(c)]}{1 - \mathbb{E}_{c \in \mathcal{D}} [\text{Pol}_{apr}(c)]} \quad (4)$$

where Pol_g is the observed nDFU in the annotations of an item c created by annotators belonging to group g .

Apunim thus measures the degree to which an annotator group contributes to overall polarization in a set of items. As demonstrated in §4; Fig. 2, our metric is robust to the number of annotators per item, opposing systematic disagreement effects, and minority groups, validating our initial motivation. We include a perturbation-based p-value estimation, which tests whether the observed *apunim* value of a group g statistically surpasses that of random groups in App. C.

5.2 How Are Humans Polarized?

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided PC dimensions can explain the systematic disagreements shown in the four datasets (§3). Statistically significant results are presented in Table 2.⁷ Note that the lack of rich annotation data prevents us from analyzing how different annotator groups interact and thus our analysis is necessarily uni-dimensional. See App. D for details on other parameters and execution times, and App. K for the full results.

⁵ $\text{Pol}_{inh}^{MC}(c) = 0 \Rightarrow \text{Pol}_{inh}(c) = 0$ due to Eq. 3, as is the case of increased k in Eq. 3.

⁶See App. H for why this observation could be explained by DICES being structurally different.

⁷The results are not necessarily contingent on a reliable p-value estimation, as non-statistically significant results have very low (< 0.01) *apunim* values. A threshold on the *apunim* values would produce similar tables.

Table 2. Statistically significant ($p < 0.05$) apunim results across the Sap and DICES datasets. Gray rows show Ethnicity and Gender groups. “Support” refers to number of annotations. Consult App. K for the full results.

Group	Apunim	Support	Group	Apunim	Support
DICES-990			DICES-350		
Ethnicity=AfricanAm	0.099	5148	Ethnicity=AfricanAm	-0.037	7627
Ethnicity=Latino	-0.063	5877	Ethnicity=Asian	0.045	6838
Ethnicity=Other	0.034	21669	Education=Other	-0.023	1841
Ethnicity=White	-0.065	10166	Sap		
Gender=Man	0.036	28817	Gender=Man	0.0907	1439
Gender=Woman	-0.027	29990	Gender=Woman	-0.2442	877
Age=GenZ	0.047	12214	POPQUORN		
Age=Millennial	-0.007	34433	Gender=Man	0.0081	2720
Age=GenX+	-0.028	12699	Race=Black	-0.1572	222
Education=College	0.006	52727	Education=College	0.0496	2388
Education=High school	-0.050	6619	Education=Graduate	-0.0808	468

Table 3. Mean inherent polarization (mean $\pm$ 2 SD) per (dataset, prompt), for Human and each LLM.

Dataset	Source	Default	Stereotype	Persona	Single	Direct
kumar	Human	0.17 $\pm$ 0.473	0.17 $\pm$ 0.473	0.17 $\pm$ 0.473	0.17 $\pm$ 0.473	0.17 $\pm$ 0.473
	llama70b	0.00 $\pm$ 0.032	0.00 $\pm$ 0.000	0.00 $\pm$ 0.045	0.00 $\pm$ 0.045	0.00 $\pm$ 0.045
	olmo32b	0.01 $\pm$ 0.137	0.03 $\pm$ 0.236	0.02 $\pm$ 0.203	0.03 $\pm$ 0.248	0.02 $\pm$ 0.201
	qwen32b	0.00 $\pm$ 0.000	0.00 $\pm$ 0.000	0.00 $\pm$ 0.000	0.00 $\pm$ 0.032	0.00 $\pm$ 0.000
	qwen7b	0.00 $\pm$ 0.095	0.01 $\pm$ 0.137	0.00 $\pm$ 0.083	0.00 $\pm$ 0.095	0.01 $\pm$ 0.109
sap	Human	0.11 $\pm$ 0.437	0.11 $\pm$ 0.437	0.11 $\pm$ 0.437	0.11 $\pm$ 0.437	0.11 $\pm$ 0.437
	llama70b	0.00 $\pm$ 0.058	0.00 $\pm$ 0.041	0.00 $\pm$ 0.058	0.00 $\pm$ 0.000	0.00 $\pm$ 0.041
	olmo32b	0.01 $\pm$ 0.136	0.01 $\pm$ 0.142	0.01 $\pm$ 0.109	0.01 $\pm$ 0.163	0.01 $\pm$ 0.148
	qwen32b	0.00 $\pm$ 0.000	0.00 $\pm$ 0.000	0.00 $\pm$ 0.000	0.00 $\pm$ 0.041	0.00 $\pm$ 0.000
	qwen7b	0.00 $\pm$ 0.000	0.00 $\pm$ 0.071	0.00 $\pm$ 0.041	0.00 $\pm$ 0.041	0.00 $\pm$ 0.041

Systematic disagreements are consistently attributed to **ethnicity** across all datasets excluding DICES-330, confirming prior work suggesting that it is a systematic source of annotator disagreement in HS tasks [16].⁸ Additionally, in both DICES-990, Sap and POPQUORN, **men** are a consistent source of systematic disagreement (positive values), while in the two former datasets **women** disagree among themselves (positive values), which is consistent with previous findings [4, 50]. None of the Kumar dimensions show statistically significant values, but this is expected, as our methodology requires at least six annotators per item to work reliably.

FINDING 4. Annotator ethnicity and gender are consistent causes of systematic disagreements, confirming prior literature on Toxicity and HS tasks.

Table 4. Cohen’s d statistics showing the quantitative difference in **nDFU** between the default and each adversarial instruction prompt for each of the groups of both datasets and across all models.

Model		Mean	Std	Min	25%	50%	75%	Max
llama70b	stereotype	0.057	0.067	-0.059	0.021	0.036	0.066	0.227
	persona	0.013	0.090	-0.102	-0.038	-0.020	0.008	0.255
	single	0.167	0.041	0.084	0.137	0.159	0.185	0.254
	direct	-0.062	0.093	-0.172	-0.113	-0.095	-0.062	0.182
olmo32b	stereotype	0.022	0.031	-0.058	0.001	0.021	0.034	0.144
	persona	0.154	0.051	0.018	0.153	0.172	0.182	0.288
	single	0.311	0.075	0.215	0.272	0.285	0.311	0.505
	direct	0.122	0.098	-0.132	0.135	0.156	0.170	0.236
qwen32b	stereotype	0.004	0.041	-0.088	-0.017	0.005	0.031	0.111
	persona	0.104	0.065	-0.073	0.098	0.122	0.138	0.216
	single	0.121	0.081	-0.091	0.124	0.149	0.165	0.241
	direct	0.092	0.062	-0.077	0.079	0.105	0.125	0.204
qwen7b	stereotype	-0.052	0.075	-0.172	-0.096	-0.073	-0.040	0.161
	persona	0.106	0.056	0.023	0.069	0.088	0.122	0.258
	single	-0.047	0.066	-0.199	-0.090	-0.063	-0.025	0.128
	direct	-0.120	0.090	-0.230	-0.170	-0.154	-0.107	0.124

6 What Polarization Patterns Do LLM Annotators Exhibit?

Experimental Procedure. We investigate four⁹ open-source LLMs with six annotators per item¹⁰ supplied with personas sampled from our human dataset. We test LLM polarization on three fronts: (1) Observed **nDFU**. (2) Overall **UnPol**. (3) Apunim values for each group. See App. I for the ablation experimental procedure, full results, model details, dataset selection, exploratory analysis, and comparison to human annotations.

Are LLMs Polarized by Default? We first examine whether persona prompting with standard instructions leads to polarized annotations. Our results show *no statistically significant attribution effects* and very small *apunim* values for all groups in all models and datasets (App. I; Table 16, 17). Unlike human annotations, we observe no **UnPol** (Table 3) and only two of the four models showing any non-zero *apunim* values (App. I; Fig. 10).

Do Instructions Influence Polarization? Since instruction prompts are known to influence the effect of personas [49], we also try four adversarial prompts progressively emphasizing the importance of the LLMs’ personas (App. L). While an ANOVA test¹¹ shows that the choice of instruction prompt does indeed affect **nDFU** scores across most models (almost all groups for OLMo-32B, half for LLaMa-70B, Qwen-7B and a single group for Qwen-32B), these effects are almost always not quantitatively significant (< 0.2 Cohen’s d), as shown in Table 4. The “single” prompt is the only prompt producing a sometimes notable (small-to-medium) quantitative effect exclusively on the OLMo-32B model, which is indeed significant as evidenced by App. I; Fig. 10. Overall, our findings support social science work which warns against generating human annotation data without substantial replication studies [23, 39].

⁸Note that the *apunim* values do not have a consistent direction across datasets, which we believe to be caused by studies using different ethnic subgroups allowing detection but distorting attribution.

⁹We used six models, but discarded two since they proved unreliable—see ablations in App. I.

¹⁰It is possible that LLMs can produce polarized patterns when scaled up to dozens or hundreds of annotators. This however is both hard to check due to exorbitant computational costs (App. I) and hard to scale up in realistic deployment scenarios.

¹¹We use a parametric test due to the abundance of observations ($N = 585, 1000$). We also use a multiple hypothesis p-value correction (specifically the “Holms-Bonferroni” method [21]).

FINDING 5. *LLMs with persona-prompting do not produce annotations with detectable **UnPol**, nor **apunim** values across any group, while only a few exhibit any multimodal annotations at all. This effect can not be reliably remedied with prompt engineering.*

Conclusion

We discovered issues regarding the presence of unattributable polarization and conflicting disagreement which render current representations of subgroup disagreement attribution suboptimal in realistic circumstances. We created a new metric that overcomes these issues and outperforms alternative methodologies in realistic annotation settings, while also quantifying the attribution of each group. By applying this metric to four subjective **NLP** datasets, we verify that ethnicity and gender seem to generally drive systematic disagreements in Toxicity and **HS** tasks. Finally, we find that persona prompting is not generally capable of unlocking consistent subgroup disagreement in subjective annotation tasks, even when using adversarial instruction prompts.

AI Use Statement

In this work we used proprietary models such as Anthropic's Claude Sonnet 5 and OpenAI's GPT-5 for partially implementing the synthetic experiments in §4 and generating parts of the library, experiment and graph-creation code. The unit-tests used in our library were similarly generated by these models, although new tests were added as edge-cases and bugs were discovered. All such tasks were executed with direct author supervision, and all relevant code files (library, experiments, graphs) have been edited manually at multiple points by the authors to ensure reproducibility, accuracy and software quality. We also used Scholar Labs to find citations for targeted claims (e.g., for known issues with persona prompting), although all such citations were manually filtered by us. Finally, we used a locally hosted Gemma-4 model (specifically the `gemma-4-E4B-it-Q4_K_M.gguf` variant) for rephrasing this manuscript in order to improve readability. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

Reproducibility Statement

We ensure that our LLM annotation study is reproducible by selecting exclusively open-source, locally hosted models and fixing all relevant seeds. An ablation study where we repeat annotations multiple times shows that our setup achieves 100% consistency across all models. In order to ensure that the LLM annotators are not influenced by minute differences in the prompts, we execute a second ablation study where we rephrase the main annotation prompt three times and compare outputs. This procedure revealed that two models were especially sensitive to such changes, which led us to exclude those models from the analysis presented in this manuscript. Information on the synthetic experiments used in §4.5 is disclosed in App. G. We analyze software design decisions, limitations and computational details in App. D and elaborate on our experimental setup in App. E, while ablations around sample sizes and sample selection for the Kumar dataset are detailed in App. H. Additionally, we include a discussion around the use of current **NLP** datasets, as well as an empirical estimation of the number of required annotators for reproducible results in App. J. All code, environment details (e.g., libraries used), data-processing scripts and our library implementing *apunim* are released anonymously: library <https://anonymous.4open.science/r/apunim-5F70>, HTML documentation <https://anonymous.4open.science/r/apunim-page>, replication code <https://anonymous.4open.science/r/aposteriori-unimodality-E8C1>.

Ethics Statement

Dual Use Concerns. The primary objective of this study is to identify systemic sources of systematic disagreements in subjective NLP datasets to facilitate the construction of more fair and reliable AI systems. However, given that our methodology allows for the quantitative attribution of specific annotation patterns to PCs, we acknowledge three primary risks inherent in this research. (1) *Discriminatory Profiling*: If the metric is applied outside of a dataset quality analysis, its ability to correlate specific opinion patterns with annotator groups could be misused for surveillance or unjust profiling. (2) *Automated Bias Amplification*: The diagnostic patterns of disagreement we detect are reflections of human systemic biases. Without human oversight, training subsequent models on these patterns risks automating and entrenching existing societal prejudices. (3) The knowledge provided by *apunim* could be *weaponized*, allowing malicious actors to strategically target weak points in moderation systems or intentionally maximize societal conflict. We stress that our metric functions strictly as a diagnostic warning signal. Any deployment in a real-world system must mandate a human-in-the-loop review, ensuring that identified patterns lead to systemic improvements (e.g., policy reform, demographic diversity initiatives) rather than merely justifying an automated system. In any case, despite the risks, we believe that the benefits of this research outweigh the risks outlined above.

Data Sourcing and Privacy. All datasets employed in this study were either publicly accessible or provided to us by the original authors with full disclosure regarding the purpose of our research. No identifiable, personal information was included in our pipeline. We do not release proprietary data provided by the authors, nor have we provided any to proprietary LLMs. Finally, we note that our data is limited to English posts from a U.S. cultural context and our results may not be representative for other cultures.

Examples Used. Our study utilizes examples of comments that may be considered divisive across certain demographics. These examples were generated by our team solely for demonstration purposes. We explicitly state that the contents of these examples and the groups studied do not necessarily reflect the views or behaviors of the groups they represent, nor do they reflect the opinions of the authors. We have partially censored explicit slurs in Table 9 in order to both mitigate uncomfortable or triggering content, while also showing enough content for the examples to be effective.

Prompts Used. This study uses adversarial prompts which explicitly instruct the model to change its outputs according to their persona prompts. The purpose of these prompts is to test the LLMs under an extreme case of prompt engineering and are not to be used for generalized usage. The use of these prompts may degrade model performance, increase the risk of stereotyping and produce misleading results.

A Acronyms

IAA	Inter-Annotator Agreement
AU	A Posteriori Unimodality
UnPol	Unattributable Polarization
nDFU	normalized Distance From Unimodality
NLP	Natural Language Processing
GPLv3	GNU General Public Licence v3
AA	African American
PC	Personal Characteristic
HS	Hate Speech

Table 5. Descriptive statistics for the number of annotations per comment, grouped by dataset.

	count	mean	std	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8313	1.1652	69	72	73	74	76
Sap et al. 2022	585	5.6359	0.7713	3	6	6	6	12
Kumar et al. 2021	2998	5.0233	0.3409	5	5	5	5	10
POPQUORN	1500	8.6573	0.9087	5	8	9	9	10

Table 6. Subgroup Counts for DICES-990 (left) and DICES-350 (right). “Support” refers to annotation count.

Category	Subgroup	Support	Category	Subgroup	Support
Gender	Woman	36050	Gender	Woman	21700
	Man	35392		Man	21350
	Nonbinary	331	Race	White	10500
	Other	330		African Am.	10150
Race	Other	26394		Asian	9100
	Asian	19885		Latino	7700
	White	12322		Multiracial	5600
	Latino	7170	Age	1) Gen. Z	19600
	African Am.	6332		2) Millennial	12600
Age	Gen. Z	14899		3) Gen. X+	10850
	Millennial	41542	Education	College +	26250
	Gen. X+	15662		High school -	14350
Education	College +	64054		Other	2450
	High school -	8049			

SD Standard Deviation

B Exploratory Analysis

The number of annotators per item for all datasets presented in §3 can be found in 6, and descriptive statistics can be found in Table 5. The DICES-990 dataset has exactly 123 annotations per item, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few items that vastly exceed the promised 5 number of annotators. We do not include these items, since they are very likely a data error. Tables 6,7,8 show the annotation counts for each PC group in each dataset.

C Statistical Estimation

Since annotation histograms may be noisy, we additionally assess whether the observed apunim value could arise by chance. To assess statistical significance, we construct a null distribution of apunim values from the randomized partitions and compare the observed apunim against this distribution using a one-sample Student-*t* test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

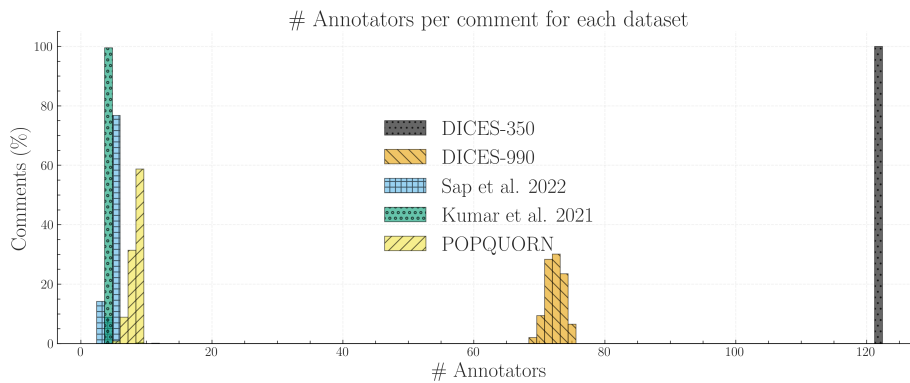


Fig. 6. We calculate the ratio of items for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few items in the Kumar dataset exceeding this threshold (see Table 5), which is likely caused by a data error.

enough annotations per PC group to estimate nDFUs, (3) the apunim values of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §3). We discuss condition (2) in App. J. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single group in principle, in practice we often want to test polarization for all subgroups of a specific dimension. Since we are simultaneously considering multiple hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holms method [21]. The test is parameterized by the confidence level of our test, which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [8].

D Software

D.1 Installation and Usage

We release the apunim package, which is a Python library that implements both the nDFU and apunim calculations, as well as its statistical test. The software is available on PyPi We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (<https://anonymous.4open.science/r/apunim-page>).

We use version 1.0.6 of the apunim library for our experiments. We use 100 random iterations for the apriori value calculation and the p-value estimation (App. C), and set a confidence level of $\alpha = 0.05$. The relative library setup is provided as a conda environment file to ensure reproducibility. The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project's repository (<https://anonymous.4open.science/r/aposteriori-unimodality-E8C1>). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely numpy, scipy, statsmodels).

Table 7. Subgroup Counts for the Sap (left) and POPQUORN (right) datasets. “Support” refers to annotation count.

Category	Subgroup	Support
Gender	Woman	252
	Man	2259
	Non-binary	786
Race	White	2168
	Black	1091
	Hisp	32
	Other	5
	MiddleEastern	1
Gender	man	1859
	woman	1412
	nonBinary	26

Category	Subgroup	Count
Gender	Woman	6470
	Man	6166
	Non-binary	350
Race	White	9754
	Black	1685
	Asian	949
	Native American	299
	Hispanic	250
	Arab American	49
Age	18-24	1545
	25-29	1236
	30-34	1494
	35-39	1547
	40-44	1142
	45-49	1091
	50-54	947
	54-59	1793
	60-64	747
	>65	1444
Education	No high school	237
	High school	3934
	College degree	6036
	Graduate degree	2481
	Other	298
Occupation	Student	995
	Homemaker	594
	Retired	1446
	Employed	6325
	Self-employed	1692
	Unemployed	1684
	Other	250

D.2 Implementation Details

By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple PC dimensions, each of which is computationally expensive. In order to lower the computational cost required, we re-use the random partitions, which are originally calculated to obtain the apriori polarization values, for the p-value calculations (App. C). This means that in our current implementation, the number of groups for estimating apriori polarization, and the number of apunim values used in the p-value calculation have to be the same.

While the library efficiently handles large numbers of annotators through vectorized operations, performance degrades significantly when processing a high volume of items, particularly when annotators belong to multiple

Table 8. Subgroup Counts for the Kumar dataset. “Support” refers to annotation count.

Category	Subgroup	Support	Category	Subgroup	Support
Gender	Female	8011	Is Trans	No	14549
	Male	6832		Yes	363
	Unknown	124		N/A	148
	Nonbinary	93	Politics	Liberal	5997
Ethnicity	White	10699		Independent	4143
	Black	1940		Conservative	3981
	Asian	862		N/A	609
	Multiracial	664		Other	330
	Hispanic	418	Is Parent	Yes	7889
	Other	247		No	7015
	Unknown	230		N/A	156
Age	Under 18	3	Tech Impact	Very negative	149
	18 - 24	1711		Somewhat negative	1411
	25 - 34	5969		Neutral	1905
	35 - 44	3785		Somewhat positive	7690
	45 - 54	1977		Very positive	3905
	55 - 64	1101	Tox Problem	Never	869
	65 or older	471		Rarely	2485
Education	N/A	43		Occasionally	5307
	No high school	71		Frequently	4441
	High School	1353		Very Frequently	1958
	College	3090	Rel Important	No	4735
	Associate’s	1676		Not very	1808
	Bachelor’s	6071		Somewhat	3548
	Master’s	2261		Very	4741
	Professional	236		N/A	228
	Doctoral	198	Seen Toxicity	True	11505
	N/A	71		False	3555
Sexual Orient.	Other	32	Targeted	False	10741
	Heterosexual	12391		True	4319
	Bisexual	1665			
	Homosexual	515			
	N/A	302			
	Other	131			

distinct groups. For instance, while experiments across all datasets in §3 complete within a minute, the Kumar dataset necessitates approximately one hour of computation using commercial hardware. Furthermore, the annotator variance analysis detailed in App. J required over ten hours. While this latter time requirement is anticipated due to the inherently demanding nature and non-standard application of our library, it demonstrates the existence of a potential scaling bottleneck. The same applies to a lesser extent for the subsampling experiments on the DICES datasets, taking collectively three hours of computation.

We hypothesize that this bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each item, for each PC dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each PC subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` package [43] for our own experiments). We also note

that the Kumar dataset also requires substantial RAM resources, estimated around 42–46GB of dedicated RAM should the whole dataset be considered (this is not a requirement for loading the sample used in this study). The CPU-bound experiments require approximately 3 hours to run, assuming a computer capable of efficient multiprocessing.

Finally, we note that in Fig. 11, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of items in DICES-990 were annotated by more than 73 annotators—see App. B).

E Apunim Design Decisions

Defining Apriori Polarization. In §5, we explained how our methodology needs to compare the polarization between the set of all annotations and those made by a specific group g . Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset—in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Fig. 4, left, we would end up subtracting the $nDFU$ s of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

Filtering Ineligible items. Because *apunim* is fundamentally designed to quantify polarization, we exclude items that exhibit no measurable polarization ($nDFU(c) = 0$). Furthermore, we also exclude items annotated exclusively by a single group (i.e., items where there are no two groups with at least three annotators each), since we require valid histograms for the non-parametric estimation of $nDFU$ (see Eq. 1). This is an unfortunately common scenario, since many NLP datasets feature only 5 – 6 annotations per item. In a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split.

F What is UnPol?

Why Does it Exist? Unlike disagreement, **UnPol** does not provide any information, but rather represents noise or uncertainty translated in terms of polarization. In order to minimize it, we need to find its cause. Since the DICES datasets are the only ones without *any* items with **UnPol** leads us to two hypotheses:

HYPOTHESIS 1 (LOW ANNOTATOR COUNT). *A low number of annotators per item may provide our model with insufficient information.*

HYPOTHESIS 2 (INACCESSIBLE GROUPS). ***UnPol** may arise from the presence of partially represented groups within the annotator pool which may go undetected if they are represented by only one or two annotators, as $nDFU$ is undefined for these groups.*

Both of these explanations are plausible and reflect known limitations of most NLP datasets. Unfortunately, we can not confirm any of these hypotheses

The fact that the DICES datasets (featuring 70+ annotators per item) exhibit zero **UnPol** in all items may initially seem to validate either of the above hypotheses. However, even when repeating the experiment using random annotator subsets (ranging from 5 to 50 annotators) and taking the maximum across ten random samples for each subset size, we still observed zero **UnPol** (App. J). This finding indicates that the DICES datasets may be structurally different. This explanation is plausible given that the annotation targets are LLM texts, which may be relatively unsubtle compared to human items, as supported by low overall polarization (Fig. 5; App B). Although a definitive answer to this question

Table 9. Sample of comments from the Sap dataset and their **UnPol**. Note that unambiguously toxic comments can be assigned with zero **UnPol**, while more nuanced cases tend to have higher values.

Content Warning: Slurs, offensive, toxic and hateful speech.	
Comment	UnPol
The minks mentioned in the headline are indeed "rising" from where they were based, but that is due to them being buried in shallow mass graves (Africa) and filling with gas that causes them to rise out. Rest assured, they're still very much dead, thankfully!	1
There should be such thing as a b!tch resistance for us women	1
There's a big difference between being a b!tch and being a woman	1
i just watched a video of @joisnotonfire being an annoying h!le on skype i'm laughing	1
n!as love to commit crime	1
I swear to God above I love ya and when ya be with this brotha, I swear I love you my bro as long as you not a n!a	0.5
I will become a cold ass b!tch because of my treatment	0.5
I would happily lay out a n!a	0.5
Shoot yourself in the head you hypersensitive fuckin b!tch.	0.5
not all women are b!tches, but all b!tches are women	0.5
"you ugly b!tch!!!" - that seems to be my real name these days"	0
"I am happy to report that drowning does kill this demonic entity. Oh holy are the waters of Chernarus!"	0
Omg, DED :)"	0
"Let them praise the name of the LORD: for his name alone is excellent; his glory is above the earth and heaven." Psalms 148:13 #Bible"	0
someone plz get this n!er some help. He's playing up like mad	0
stop talking you fucking n!er !	0

eludes us due to the lack of other datasets with high annotator counts per item, the indication that the DICES datasets may not feature similar polarization patterns is noteworthy.

How Does it Look Like? Table 9 shows a random sample of items taken from the Sap dataset and their respective **UnPol** values (§4). Note that explicitly toxic or racist items tend to have **UnPol** values of zero, alongside clearly unproblematic items. Inversely, more ambiguous and less comprehensible items tend to have larger **UnPol** values. This pattern may suggest that **UnPol** may be tied to the overall ambiguity of the statements, especially in cases where items may either be interpreted as very toxic, or non-toxic, which agrees with the overall goal of **nDFU** (§2). We include the **UnPol** values for each individual item for all four datasets in the supplementary material.

G Synthetic Experiments

This appendix details the synthetic experiments used to illustrate the functionality and behavior of **nDFU**. In all simulations, the bin size is set to $N_{bins} = 10$.

Disagreement vs. Polarization. To explore the relationship between annotation polarization and disagreement in Fig. 1, we constructed a 2×2 matrix of synthetic datasets, each drawing from a pool of $N_{annotators} = 100$. Polarization is simulated by defining a minority group, which constitutes 20% of the total annotators ($n_{special} = 20$). These minority annotators are assigned a targeted mean shift ($mean_shift = 1$) from the base distribution. The base variance for these

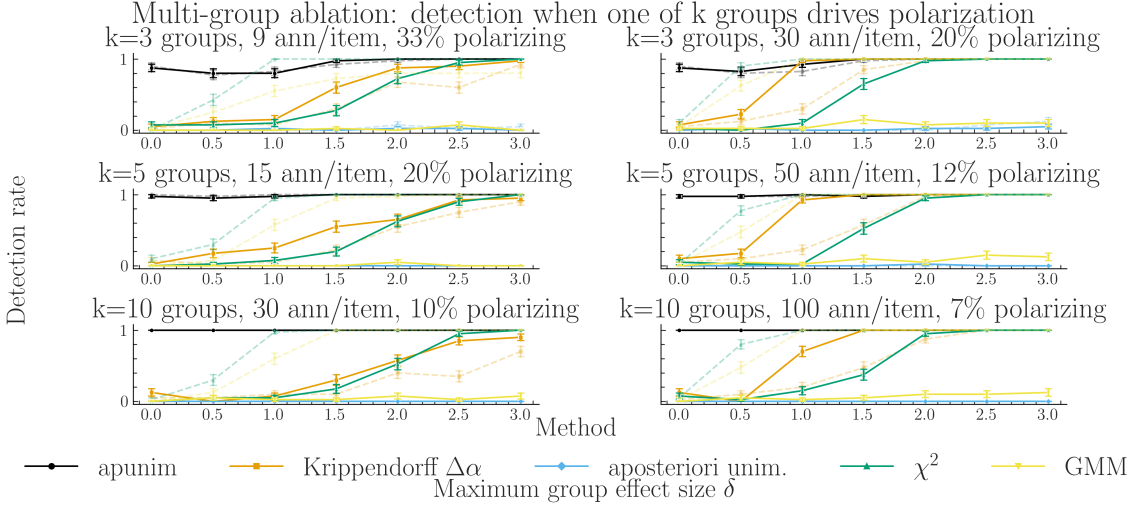


Fig. 7. Polarization group attribution on synthetic distributions. Bars indicate standard error. The setup is identical to the one used for Fig. 2, although in these experiments we consider multiple k annotator groups for each item. Dashed lines show the results of a simplified polarization setup, where polarization always has the same direction.

distributions is set at variance = 0.3. Different types of distributions (unimodal, bimodal, multimodal, uniform) are achieved by controlling the base_means and applying variance multipliers (e.g., variance $\times 3$ for multimodal).

Single-Item Polarization. We simulate the distribution of toxicity annotations in Fig. 4, left, across two demographic groups (Men and Women). The distribution of Men’s scores is modeled using a truncated normal distribution with loc = 2 and scale = 1.3, while Women’s scores are modeled with loc = 8 and scale = 1.3. The combined dataset, representing all annotations, is used to demonstrate the impact of population weighting on the overall nDFU score. The sample size for each group in this visualization is $N_{annotators} = 50$.

Discussion-Level Polarization. We examine nDFU scores for three specific textual examples presented in Fig. 3. For the individual items, $N_{annotators} = 200$ for each group. In the first case (misogynist item), Men’s scores are modeled with loc = 6, while Women’s scores are modeled with loc = 2. In the second case (misandrism item), the roles are reversed. The third visualization combines distributions for men and women derived from the respective single-item distributions.

H Experiment Ablations

Why Do DICES not Exhibit UnPol? In order to investigate whether the numerous annotators per items can explain the lack of UnPol in the DICES datasets (§4) we gradually subsample the annotators for each item randomly with replacement, starting from five annotators per item, and scaling up to the full complement of annotators in the dataset with a step of two. For each such step, we choose ten annotator samples and compute the maximum UnPol across all runs. Even with annotator counts similar to the Kumar and DICES datasets, UnPol remains zero, suggesting that the absence of UnPol can not be attributed to annotator number alone—rather, it is plausibly a pattern attributable to the DICES datasets themselves. This kind of analysis may be used in the exploratory phase of dataset selection in order to filter out datasets without enough group-specific subjectivity.

Does Apunim Work for More than Two Groups? We repeat the experiment comparing apunim to prior methodologies shown in Fig. 2 but generalized across multiple groups. As shown in Fig. 7, apunim stays competitive even in the presence of multiple annotator groups. Note the increased efficiency of many methods when polarization is unidirectional. This pattern indicates that the challenges of bidirectional polarization severely limiting prior approaches may be generalizable to more than two groups.

Table 10. Comparison of Apunim values across different sample sizes (1k, 3k, 10k, 30k) for the Kumar dataset.

		1000	3000	10000	30000
PC					
Age	1) 18 - 24	0.014	-0.017	-0.037	-0.016
	2) 25 - 34	-0.003	-0.000	0.000	0.000
	3) 35 - 44	0.000	-0.011	-0.001	-0.001
	4) 45 - 54	0.022	0.033	0.013	0.003
	5) 55 - 64	0.000	0.000	-0.057	0.001
	6) 65 or older	—	—	—	0.948
Education	2) High School graduate	0.000	0.000	-0.005	0.014
	3) College, no degree	0.010	0.014	0.010	0.006
	4) Associate degree	0.000	0.000	0.005	0.009
	5) Bachelor's degree	0.001	-0.000	-0.001	-0.000
	6) Master's degree	0.000	-0.006	0.004	-0.000
Ethnicity	Asian	-0.000	0.000	0.011	0.007
	Black	0.000	-0.024	-0.004	-0.012
	Hispanic	—	—	0.000	-0.099
	Multiracial	0.000	0.000	0.000	0.000
	Other	—	—	—	-0.306
	White	0.000	-0.000	-0.000	0.000
Gender	Female	0.001	0.000	0.002	0.001
	Male	0.001	-0.004	-0.002	-0.001
Has Been Targeted	False	0.001	0.000	-0.000	0.000
	True	-0.009	-0.015	-0.003	-0.002
Is Parent	No	-0.002	0.002	0.001	0.001
	Prefer not to say	—	—	1.000	0.000
	Yes	-0.000	0.000	-0.000	0.000
Is Transgender	No	0.000	0.000	0.000	0.000
Political Affiliation	Conservative	-0.013	0.003	-0.003	-0.002
	Independent	0.000	-0.001	0.001	0.002
	Liberal	-0.001	0.006	0.001	0.001
	Other	—	0.000	0.000	0.000

Continued on next page

Table 10. Comparison of Apunim values across different sample sizes (1k, 3k, 10k, 30k) for the Kumar dataset.

		1000	3000	10000	30000
PC					
	Prefer not to say	—	—	0.000	0.022
Religion Important	1) No	0.002	0.004	0.002	0.001
	2) Not very	0.000	-0.006	0.010	0.006
	3) Somewhat	0.000	0.002	0.003	-0.002
	4) Very	0.000	-0.003	-0.003	-0.002
Seen Toxicity	False	-0.007	0.002	-0.001	-0.001
	True	-0.001	0.000	0.001	0.000
Sexual Orientation	Bisexual	-0.068	-0.029	-0.040	-0.013
	Heterosexual	-0.001	-0.000	-0.000	0.000
	Homosexual	—	—	0.000	0.053
	Prefer not to say	—	—	—	0.370
Technology Impact	2) Somewhat negative	0.000	0.000	0.000	0.003
	3) Neutral	0.124	0.037	-0.014	0.000
	4) Somewhat positive	0.004	0.001	-0.000	0.001
	5) Very positive	-0.015	-0.004	-0.004	-0.003
Toxicity Problem	1) Never	0.000	0.000	-0.037	-0.081
	2) Rarely	0.000	0.000	0.003	-0.001
	3) Occasionally	0.000	-0.004	-0.003	-0.001
	4) Frequently	0.000	0.000	-0.001	-0.000
	5) Very Frequently	0.000	-0.002	0.030	0.015

How Does Dataset Size Influence Apunim? Since our approach was designed for the small-scale datasets common in NLP, we were cautious when testing it on the massive Kumar dataset. Because of the data volume, we analyzed various smaller samples (1,000, 3,000, 10,000, and 30,000 items), shown in Table 10. Sample size does not seem to have a consistent effect on apunim values, other than large splits uncovering significant polarization effects in sparsely populated groups (e.g., 65+ yo. in Age, “Prefer Not to Say” in *Sexual Orientation*, “Never”, “Very Frequently” in *Toxicity Problem*). The apunim direction stays broadly the same. We also test whether the results in Table 2 are an artifact of the dataset split, because of the small sample size used in the study. We repeat the apunim experiment with ten different subsets of the same dataset and record their apunim values, shown in Fig. 8. We find that most apunim values are stable, save for “Never” in *Toxicity Problem*, “Heterosexual” in *Sexual Orientation* and “Neutral” in *Technology Impact*.

How Does Sample Selection Influence Apunim? Astute readers may notice that some variability does exist across differently sized samples for the same examined groups, where for example the sign of the apunim value may switch. We note that the statistically significant values (bold rows) change signs only when the apunim value approaches zero in the largest samples. Thus, the results are not inverted, but rather pushed towards zero. In order to verify that this variability is caused by the sample size, and not by different samples being drawn, we execute a further ablation study.

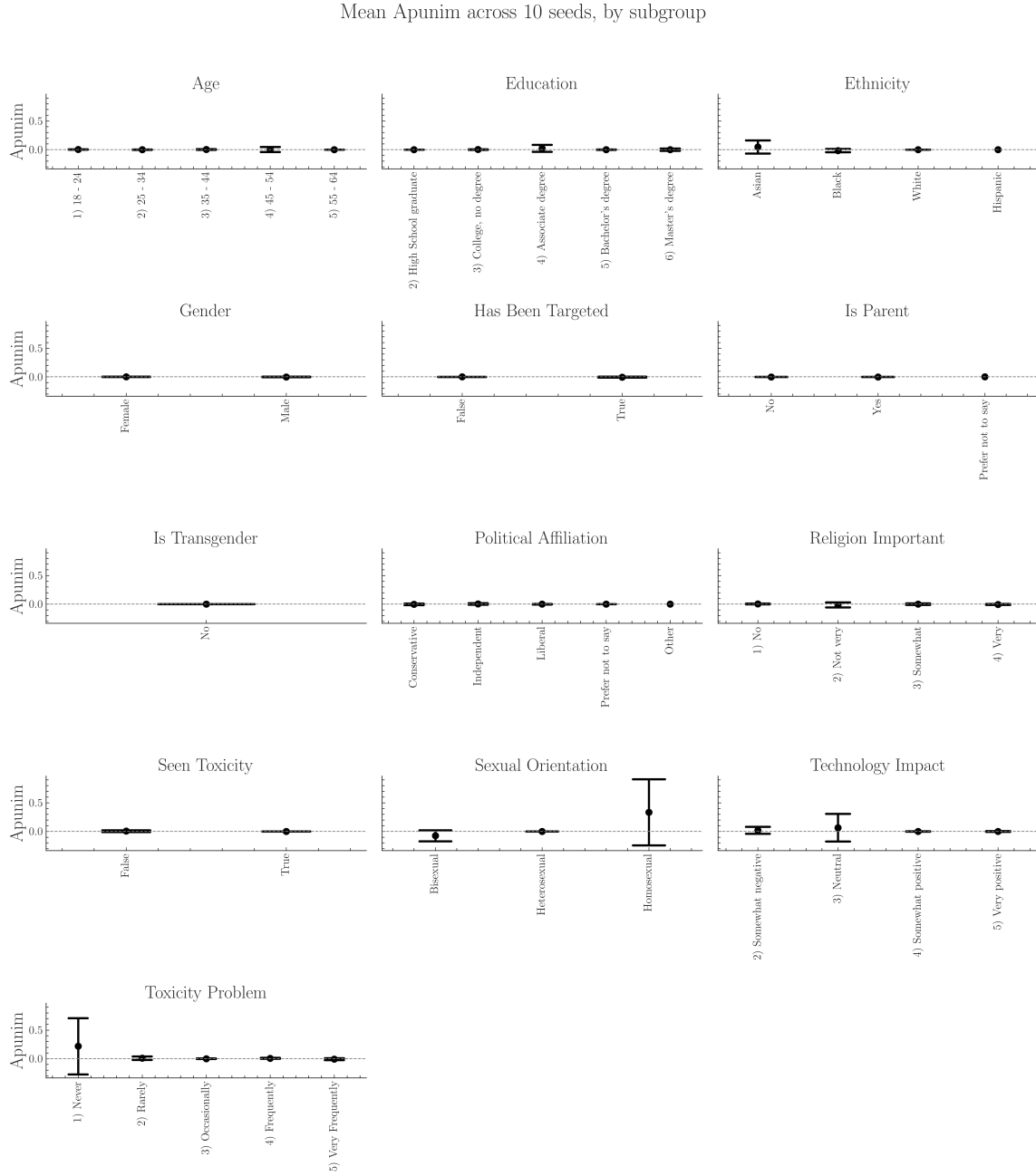


Fig. 8. Apunim values on the Kumar dataset ($n_{samples} = 1000$) across ten runs with different subsets. Black bars indicate two Standard Deviations (SDs). PC dimensions with many groups are more volatile, although all dimensions broadly keep the same apunim sign across subsets.

Specifically, we execute the apunim study on the 3000-sized sample of the dataset, using different subsets of the Kumar dataset. The results can be found in Fig. 8, where we note that variability is not an issue, except for a couple of cases. These cases seem to represent sparse annotator groups, where missing a few items may genuinely change the value of the apunim metric. These groups however are largely not detected by apunim as significant either way (see Table 2).

I LLM Annotation Procedure

I.1 Models Used

We use only open-source models, as proprietary models such as OpenAI’s GPT-5 yield much less reproducible results [5]. Furthermore, the use of open-source models requires much fewer resources, enabling research on this task with a less demanding budget, as well as scalable deployment in real-world settings [39]. It is also worth noting that since the datasets used are not publicly licensed, we can not risk leaking the data contained within to proprietary models. Generally, the use of local models protects the personal data of the users contained in our datasets, ensures that our research can be replicated in domains where anonymity must be maintained, and renders our work compatible with existing legislation [11, 12]. Table 11 shows details on the LLMs used for this study. We specifically use 4-bit quantized versions from the unsloth huggingface repository.

Table 11. LLMs used in this study.

Family	Version	#Params
OLMo [18]	2 (0325)	32B
Qwen [52]	2.5	32B
OLMo [34]	3	7B
Qwen [52]	2.5	7B
LLaMa [17]	3.1	8B

I.2 Experimental Setup

Since each of the datasets in §3 was annotated using different annotation guidelines and targets we use different instruction prompts for each dataset. The “default” annotation prompt is directly extracted from the studies in the case of Sap and Kumar, and inferred from the instruction guidelines used in the DICES datasets with minor modifications. We use standard persona prompting procedures sampled randomly from the available PC dimensions in each respective dataset. Six LLM annotators are selected for each item and model.

The prompts used in the ablation study were provided by a GPT-5 and a local Gemma-4 (quantized) model by providing the original prompt and instructing it to produce four variants. The tested prompts use different methods to restrict the model’s behavior. The initial prompts (“*Persona*”) require the model to adopt a persona’s viewpoint to judge the comment. Other prompts (“*direct*”) treat the persona’s characteristics as an absolute, non-negotiable rule that overrides standard moderation guidelines. Another approach (“*single*”) compels the model to perform a persona-based check before submitting its final rating. The most aggressive prompts (“*stereotype*”) force the model into extreme decision spaces. The full main and ablation annotation prompts can be found in App. L.

We use a single Quadro RTX 6000 GPU for tasks requiring small (7-8B) models and two such GPUs for the rest. The annotation task, including ablations, requires approximately 300 hours of computation to complete. We use the transformers library to load and manage the LLMs. We release the relevant code in the supplementary material. Our

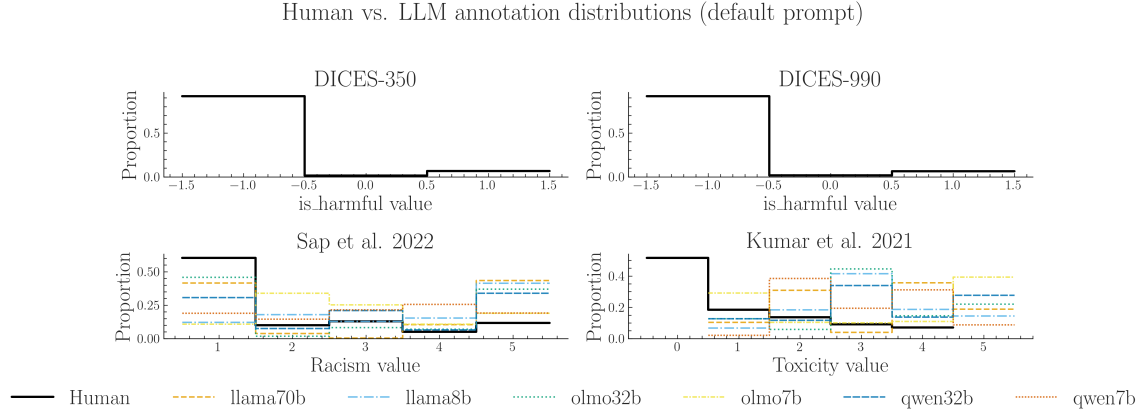


Fig. 9. Annotation frequencies for humans and each LLM in the primary annotation task using the default prompt.

codebase is platform-independent except for three high-level bash scripts and a full environment file is included for reproduction.

I.3 Exploratory Analysis

Fig. 9 shows the different annotation frequencies of LLMs and human annotators in the same dataset split. Across all four datasets, we observe a consistent mismatch between human and LLM annotation distributions. For the DICES-350 and DICES-990 harmfulness tasks, human annotations are heavily concentrated at the lowest harmfulness value, whereas most LLMs assign substantially more probability to moderate and high harmfulness values. A similar pattern appears in the Sap et al. [42] racism dataset: human judgments are concentrated at low racism scores, while LLMs produce broader distributions with considerably more mass at higher scores, particularly values 4–5. The [25] toxicity results show the same general trend, with humans assigning most probability to zero toxicity while LLMs distribute their predictions across higher toxicity levels, often peaking around values 2–4. Importantly, the magnitude of this discrepancy varies across models; some LLMs are closer to the human distribution, while others show substantially stronger shifts toward higher scores.

We also investigate whether LLM annotations include any **UnPol**. Table 3 shows that **UnPol** is significantly smaller than in human annotations. Adversarial prompts seem to consistently increase it, although it does not approach human levels. It is also notable that as with LLM apunim values (next section), **UnPol** seems to be primarily model-dependent.

I.4 Annotation Ablations

We examine the following conditions for selecting models and datasets for our LLM annotation experiment. All ablation experiments use a sampled 20% fraction of the full LLM annotation dataset used in the main text.

The LLMs should be consistent given the same prompt. While we have fixed all relevant seeds, it is possible that generation deviates across repeated model prompting given the same input [5]. We check this condition by providing the model with the main annotation prompt (see App. L) five times and find that generation is indeed deterministic under our setup for all models (Table 12).

Table 12. Consistency (Krippendorff's α , ordinal) of each model with itself across repeated runs of the same (default) prompt.

Dataset	Model	Runs	Items	Krippendorff's alpha
Sap et al. 2022	llama70b	5	180	1.0000
Sap et al. 2022	llama8b	5	180	1.0000
Sap et al. 2022	olmo32b	5	178	1.0000
Sap et al. 2022	olmo7b	5	172	1.0000
Sap et al. 2022	qwen32b	5	180	1.0000
Sap et al. 2022	qwen7b	5	180	1.0000
Kumar et al. 2021	llama70b	5	600	1.0000
Kumar et al. 2021	llama8b	5	600	1.0000
Kumar et al. 2021	olmo32b	5	600	1.0000
Kumar et al. 2021	olmo7b	5	567	1.0000
Kumar et al. 2021	qwen32b	5	600	1.0000
Kumar et al. 2021	qwen7b	5	589	1.0000

Table 13. Consistency (Krippendorff's α , ordinal) of each model with itself across the three paraphrased prompt variants.

Dataset	Model	Variants	Items	Krippendorff's alpha
Sap et al. 2022	llama70b	3	180	0.8829
Sap et al. 2022	llama8b	3	180	0.6717
Sap et al. 2022	olmo32b	3	180	0.9086
Sap et al. 2022	olmo7b	3	180	0.7090
Sap et al. 2022	qwen32b	3	180	0.9199
Sap et al. 2022	qwen7b	3	180	0.7993
Kumar et al. 2021	llama70b	3	600	0.8406
Kumar et al. 2021	llama8b	3	600	0.3988
Kumar et al. 2021	olmo32b	3	600	0.7827
Kumar et al. 2021	olmo7b	3	593	0.3897
Kumar et al. 2021	qwen32b	3	600	0.9457
Kumar et al. 2021	qwen7b	3	600	0.5578

Table 14. Consistency (Krippendorff's α , ordinal) between LLMs given the same (default) prompt, per dataset.

Dataset	Prompt	Models	Items	Krippendorff's alpha
Sap et al. 2022	default	6	3510	0.6556
Kumar et al. 2021	default	6	7741	0.4519

Table 15. Consistency (Krippendorff's α , ordinal) between LLMs given the same (default) prompt, per dataset, excluding the following models: llama8b, olmo7b.

Dataset	Prompt	Models	Items	Krippendorff's alpha
Sap et al. 2022	default	4	3510	0.7594
Kumar et al. 2021	default	4	6000	0.5999

The LLMs should be prompt insensitive. Small changes in the prompt have been shown to drastically affect LLM output in some tasks depending on the experimental setup [38]. In order to verify whether any models are unreliable under our setup, we create two alternative formulations of each prompt without a change in substance using a GPT-5 model (free version), whose outputs we manually verified (see App. L). Table 13 shows that while most LLMs are relatively robust to changes in formulation, the smaller ones (OLMo-7B and LLaMa-8B) are not. They are thus excluded from the annotation study.

The annotation guidelines should be unambiguous. The studies that provided the datasets used in our work (see §3) do not include LLM annotation, nor equivalent LLM prompts, which meant we had to generate them from the human annotation guidelines ourselves. This was not a problem in the case of the Sap and Kumar datasets, since the annotation guidelines were short, simple and required minimal edits to be transformed into LLM annotation guidelines. However, the DICES datasets use poorly documented guidelines and the target label we used is an aggregation of multiple annotation tasks. For these reasons the prompt creation is non-trivial and prone to errors. Indeed, Table 14 shows that inter-model agreement is extremely low for the DICES datasets, even when we remove the sensitive models (Table 15). Given that the DICES datasets are (1) problematic w.r.t. inherent polarization (see App. G), (2) conditioned on several dozen annotations per item (3) difficult to find reliable annotation prompts for LLMs, we decided to exclude them from the annotation study.

I.5 LLM Annotation Results

Overall Polarization. Fig. 10 shows the overall polarization patterns when using the different adversarial instructions. The Qwen-23B and LLaMa-70B models consistently produce no polarization, while the persona-focused prompt seems to elicit some varied polarization, even though it does not translate to higher apunim values. This pattern would indicate that the persona-focused prompt likely adds more overall variance to the persona-conditioned annotations, even though that variance is not consistent.

Apunim Analysis. Tables 16 and 17 show the apunim results for the LLM annotation task on the Kumar and Sap datasets respectively. Apunim values are largely model-dependent, but no other concrete pattern can be discerned.

Table 16. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
Age	1) 18 - 24	llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	—	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	2) 25 - 34	llama70b	—	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	3) 35 - 44	llama70b	0.000	0.000	0.000	0.000	—
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	—	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	4) 45 - 54	llama70b	0.000	0.000	0.000	-0.096	0.000
		olmo32b	-0.005	0.017	-0.021	0.018	-0.048
		qwen32b	0.000	0.000	0.000	0.000	0.000

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Table 16. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
	5) 55 - 64	qwen7b	-0.007	0.000	0.000	-0.043	0.000
		llama70b	0.000	0.000	—	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
	6) 65 or older	qwen7b	0.000	0.000	-0.000	0.000	0.000
		llama70b	0.000	0.000	0.000	0.135	0.000
		olmo32b	-0.029	0.002	-0.023	0.006	-0.033
		qwen32b	0.166	—	0.000	0.000	0.000
	Prefer not to say	qwen7b	0.031	-0.031	-0.012	-0.001	0.000
		llama70b	0.000	0.000	0.000	-0.130	0.000
		olmo32b	-0.016	-0.019	-0.009	-0.043	-0.003
		qwen32b	-0.222	0.000	0.000	0.000	0.000
	Under 18	qwen7b	-0.033	-0.028	-0.015	0.025	0.000
		llama70b	0.000	—	0.000	0.000	—
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	—	0.000	0.000	—
Education	1) No high school	qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	—	0.000	0.000	—
		qwen32b	0.000	0.000	0.000	0.000	0.000
	2) High School graduate	qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	0.000	—	0.000	—
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	—	—	0.000	0.000	0.000
	3) College, no degree	qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	—	—	—	—
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	-0.000	0.000	0.000
	4) Associate degree	qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	—	0.000	0.000	—	0.000
	5) Bachelor's degree	qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	—	0.000	0.000
	6) Master's degree	qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	—	0.000	0.000	0.000	—
		olmo32b	—	0.000	0.000	0.000	0.000
		qwen32b	—	—	0.000	0.000	—
	7) Professional degree	qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
	8) Doctoral degree	qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	0.000	—	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	—	0.000	0.000	—
Ethnicity	Asian	qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.069	0.074	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	—	0.000	0.000	0.000	0.000
	Prefer not to say	qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	—	0.000	0.000	0.000
	Other	qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	0.000	0.000	0.000	—
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	—	0.000	0.000	0.000	0.000

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Table 16. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
Gender	Black	olmo32b	0.000	0.012	-0.017	0.008	0.000
		qwen32b	0.000	-0.083	0.000	—	0.000
		qwen7b	0.000	-0.023	-0.012	-0.006	-0.019
		llama70b	0.000	0.000	0.000	0.000	0.000
	Hispanic	olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	0.000	0.000	0.000	0.000
	Multiracial	olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	—	0.000	0.000	0.000
	Other	olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	0.000	0.000	0.000	0.000
	Unknown	olmo32b	-0.190	-0.194	0.000	0.000	0.000
		qwen32b	0.000	0.017	-0.007	0.057	0.000
		qwen7b	0.000	-0.030	0.000	-0.032	0.000
		llama70b	0.000	-0.032	-0.022	-0.006	-0.011
	White	olmo32b	0.000	0.000	—	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
		llama70b	0.000	0.000	-0.000	0.000	0.000
Has Been Targeted	Female	olmo32b	-0.010	0.015	-0.035	0.109	-0.003
		qwen32b	0.036	0.002	0.016	-0.004	-0.017
		qwen7b	-0.109	-0.046	-0.012	-0.050	0.000
		llama70b	0.035	0.011	0.024	0.005	-0.005
	Male	olmo32b	0.003	-0.017	0.052	0.005	-0.063
		qwen32b	0.017	-0.033	0.012	0.033	0.012
		qwen7b	0.017	-0.084	-0.042	0.001	0.000
		llama70b	-0.005	0.029	0.006	-0.014	0.024
	Nonbinary	olmo32b	-0.025	-0.039	0.044	0.031	0.078
		qwen32b	-0.003	0.018	-0.013	-0.008	-0.003
		qwen7b	0.000	0.084	0.059	-0.015	0.000
		llama70b	0.020	-0.024	-0.007	-0.003	0.015
	Unknown	olmo32b	0.011	0.037	-0.011	-0.027	0.022
		qwen32b	-0.012	-0.038	-0.024	-0.028	0.008
		qwen7b	0.060	-0.026	0.012	0.019	0.000
		llama70b	-0.044	0.003	0.047	0.029	-0.005
Is Parent	False	olmo32b	0.002	-0.001	-0.024	-0.026	-0.009
		qwen32b	0.004	0.023	0.025	0.017	-0.003
		qwen7b	0.013	0.002	0.000	-0.032	-0.021
		llama70b	0.008	-0.005	-0.004	0.003	0.008
	True	olmo32b	-0.057	-0.001	0.043	0.023	-0.021
		qwen32b	-0.021	-0.021	-0.016	-0.001	0.005
		qwen7b	-0.041	0.036	-0.006	0.014	0.024
		llama70b	-0.022	-0.015	-0.001	-0.005	-0.015
Prefer not to say	No	olmo32b	-0.010	0.011	-0.057	-0.034	-0.009
		qwen32b	-0.016	-0.017	-0.017	0.019	0.016
		qwen7b	0.045	0.024	-0.006	-0.023	-0.025
		llama70b	-0.013	0.018	-0.024	-0.008	0.012
	Prefer not to say	olmo32b	-0.021	-0.003	0.031	0.018	-0.035
		qwen32b	0.008	0.045	0.028	-0.000	-0.025

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Table 16. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct	
	Yes	qwen32b	-0.017	-0.047	0.111	0.036	0.035	
		qwen7b	0.013	-0.018	-0.010	-0.006	-0.021	
		llama70b	0.024	-0.015	0.009	0.013	0.027	
		olmo32b	0.023	-0.018	-0.014	-0.013	0.005	
		qwen32b	0.014	0.073	-0.037	-0.016	-0.001	
		qwen7b	0.018	-0.028	-0.002	-0.011	-0.028	
Is Transgender	No	llama70b	0.020	0.033	0.030	0.041	0.005	
		olmo32b	0.014	0.025	-0.030	0.028	0.036	
		qwen32b	0.029	0.030	0.012	-0.041	-0.024	
		qwen7b	-0.019	-0.008	-0.012	0.034	-0.001	
	Prefer not to say	llama70b	-0.025	-0.037	-0.047	-0.048	0.043	
		olmo32b	-0.005	-0.024	0.025	-0.012	0.020	
		qwen32b	0.031	-0.046	0.021	-0.020	-0.026	
		qwen7b	0.039	0.002	0.017	-0.006	-0.014	
	Yes	llama70b	0.016	0.040	-0.016	-0.013	-0.064	
		olmo32b	0.009	0.018	-0.035	-0.005	-0.024	
		qwen32b	-0.103	0.029	-0.019	0.001	0.029	
		qwen7b	-0.039	0.003	0.024	-0.027	0.029	
Political Affiliation	Conservative	llama70b	0.000	0.000	0.034	-0.018	-0.017	
		olmo32b	-0.003	0.019	0.013	0.015	0.025	
		qwen32b	-0.084	0.000	-0.020	0.000	0.000	
		qwen7b	-0.025	-0.018	0.028	0.010	0.013	
	Independent	llama70b	0.000	0.000	0.047	-0.031	0.000	
		olmo32b	0.009	-0.018	0.019	0.019	0.001	
		qwen32b	-0.054	0.000	0.000	0.000	-0.038	
		qwen7b	-0.016	0.004	0.000	0.006	0.019	
	Liberal	llama70b	0.000	0.000	-0.075	0.026	0.000	
		olmo32b	0.000	0.009	-0.021	-0.012	-0.013	
		qwen32b	0.037	0.000	-0.010	0.000	-0.005	
		qwen7b	-0.004	0.023	-0.015	0.006	0.034	
	Other	llama70b	0.000	0.000	0.000	-0.041	0.071	
		olmo32b	-0.010	-0.005	-0.019	-0.023	0.019	
		qwen32b	0.000	0.000	0.000	0.000	0.000	
		qwen7b	0.031	-0.005	-0.003	0.009	-0.006	
	Prefer not to say	llama70b	0.000	0.000	0.000	0.048	0.000	
		olmo32b	-0.006	0.023	0.009	-0.013	-0.001	
		qwen32b	0.061	0.000	0.000	0.000	0.059	
		qwen7b	0.015	0.017	0.019	0.000	-0.015	
	Religion Important	1) No	llama70b	-0.007	0.017	-0.014	0.009	-0.057
			olmo32b	-0.018	0.009	-0.020	0.006	-0.015
			qwen32b	-0.014	0.074	-0.067	-0.017	0.000
			qwen7b	-0.078	-0.013	-0.006	-0.007	-0.003
2) Not very		llama70b	-0.021	-0.020	0.000	-0.029	0.000	
		olmo32b	-0.021	0.007	0.020	0.008	0.008	
		qwen32b	0.000	0.000	0.059	0.000	0.096	
		qwen7b	0.011	-0.005	0.021	0.020	0.026	
3) Somewhat		llama70b	0.000	0.000	0.000	0.000	0.000	
		olmo32b	0.017	0.028	-0.023	-0.012	0.028	
		qwen32b	0.000	0.000	0.000	0.000	0.000	
		qwen7b	-0.026	-0.003	-0.006	0.017	0.000	
4) Very		llama70b	0.006	0.005	-0.048	0.000	0.017	
		olmo32b	0.011	0.022	0.013	0.018	0.020	
		qwen32b	0.000	-0.038	0.038	0.000	0.000	
		qwen7b	-0.005	-0.021	-0.025	-0.054	-0.015	
Prefer not to say		llama70b	0.015	0.049	0.136	-0.066	0.000	
		olmo32b	-0.003	-0.009	-0.004	0.002	-0.008	
		qwen32b	0.092	0.000	-0.130	-0.012	-0.017	
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Table 16. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
Seen Toxicity	False	qwen7b	0.013	0.007	0.008	0.007	-0.005
		llama70b	0.027	0.038	0.021	0.007	0.007
		olmo32b	-0.012	0.002	-0.013	0.006	-0.003
		qwen32b	0.013	-0.055	0.003	-0.069	-0.066
	True	qwen7b	-0.005	-0.012	0.011	-0.007	0.039
		llama70b	-0.033	-0.017	-0.002	0.005	0.085
		olmo32b	-0.011	0.029	-0.008	0.002	-0.004
		qwen32b	-0.004	0.013	0.015	0.016	0.052
	Sexual Orientation	qwen7b	0.013	0.012	-0.011	-0.016	-0.032
		llama70b	0.000	0.000	0.000	-0.002	0.050
		olmo32b	-0.026	0.020	-0.004	-0.032	0.029
		qwen32b	0.000	0.047	0.000	0.083	-0.018
Technology Impact	Bisexual	qwen7b	-0.018	0.004	0.013	-0.007	0.027
		llama70b	0.000	0.000	0.000	0.061	0.000
		olmo32b	-0.023	0.016	-0.005	0.026	0.005
		qwen32b	-0.033	-0.057	0.000	-0.021	-0.049
	Heterosexual	qwen7b	-0.004	-0.004	0.000	0.023	0.002
		llama70b	0.000	0.000	0.000	-0.038	-0.026
		olmo32b	0.039	0.000	-0.009	-0.005	0.000
		qwen32b	0.000	0.000	-0.029	-0.079	0.000
	Homosexual	qwen7b	0.009	-0.018	-0.030	0.023	0.030
		llama70b	0.000	0.000	0.000	-0.032	0.000
		olmo32b	-0.002	-0.023	0.004	0.003	0.017
		qwen32b	-0.061	0.000	0.000	0.000	0.048
	Other	qwen7b	-0.025	-0.004	-0.002	-0.032	0.012
		llama70b	0.000	0.000	0.000	-0.028	-0.024
		olmo32b	-0.019	0.011	0.010	-0.020	0.016
		qwen32b	0.000	0.085	0.036	-0.025	0.060
	Prefer not to say	qwen7b	0.024	-0.018	-0.027	-0.009	0.042
		llama70b	0.085	-0.010	0.000	0.000	0.000
		olmo32b	0.000	-0.013	0.000	0.000	0.016
		qwen32b	0.000	0.000	-0.061	0.000	-0.029
	1) Very negative	qwen7b	0.000	0.000	0.000	0.008	0.000
		llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.010	-0.026	-0.008	0.000	-0.022
		qwen32b	0.000	0.000	0.000	0.000	-0.013
	2) Somewhat negative	qwen7b	0.000	0.000	0.000	0.010	0.000
		llama70b	0.000	0.000	-0.042	0.050	0.000
		olmo32b	-0.028	0.002	0.019	0.017	0.024
		qwen32b	-0.346	-0.017	0.000	0.000	-0.067
	3) Neutral	qwen7b	0.000	-0.002	0.026	-0.021	-0.030
		llama70b	-0.017	0.000	0.059	-0.019	0.000
		olmo32b	0.007	-0.009	-0.009	-0.035	-0.001
		qwen32b	0.139	0.081	0.000	0.000	0.000
	4) Somewhat positive	qwen7b	0.005	0.011	-0.008	0.000	-0.023
		llama70b	-0.038	-0.009	0.000	-0.006	0.000
		olmo32b	0.025	0.065	0.016	0.001	-0.024
		qwen32b	0.000	0.195	0.107	0.000	-0.014
	5) Very positive	qwen7b	0.012	0.011	0.027	-0.022	0.015
		llama70b	-0.049	0.001	0.062	-0.025	0.000
		olmo32b	-0.010	-0.004	-0.010	-0.014	0.029
		qwen32b	0.000	0.000	-0.007	-0.015	0.000
Toxicity Problem	1) Never	qwen7b	-0.015	0.018	-0.018	0.008	-0.034
		llama70b	0.000	0.000	0.078	0.027	-0.011
		olmo32b	-0.013	0.009	0.001	-0.012	-0.012
		qwen32b	0.000	-0.043	0.000	0.042	0.000
	2) Rarely	qwen7b	-0.003	-0.004	0.017	0.022	0.000
		llama70b	-0.049	0.001	0.062	-0.025	0.000
		olmo32b	-0.010	-0.004	-0.010	-0.014	0.029
		qwen32b	0.000	0.000	-0.007	-0.015	0.000

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Table 16. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
	3) Occasionally	llama70b	-0.009	0.000	0.000	0.000	0.000
		olmo32b	0.008	0.019	0.004	0.004	-0.020
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.012	-0.019	0.012	-0.018	0.031
	4) Frequently	llama70b	-0.008	0.046	0.000	0.023	0.000
		olmo32b	-0.001	-0.035	-0.028	-0.005	-0.017
		qwen32b	0.000	0.000	-0.020	-0.043	0.000
		qwen7b	-0.016	-0.013	0.021	-0.017	0.010
	5) Very Frequently	llama70b	0.050	-0.046	-0.051	-0.030	-0.032
		olmo32b	-0.006	-0.022	-0.005	-0.001	0.004
		qwen32b	0.000	-0.018	-0.012	-0.031	0.000
		qwen7b	-0.010	0.010	-0.014	0.010	0.018

Table 17. Aposteriori unimodality results for the LLM annotations of the Sap et al. 2022 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
Age	1) Gen. X+	llama70b	0.011	0.020	-0.078	-0.057	0.022
		olmo32b	-0.054	0.038	0.024	-0.002	0.009
		qwen32b	0.008	0.000	-0.004	-0.076	0.043
		qwen7b	-0.033	0.021	0.013	0.007	0.036
	2) Millennial	llama70b	-0.069	0.016	-0.027	0.027	0.027
		olmo32b	0.011	0.005	0.008	-0.022	0.020
		qwen32b	-0.028	-0.041	0.016	0.076	0.068
		qwen7b	0.023	0.002	-0.045	0.013	-0.053
	3) Gen. Z	llama70b	0.197	0.007	0.018	-0.012	-0.041
		olmo32b	0.001	0.023	-0.016	0.015	-0.026
		qwen32b	0.052	-0.035	-0.017	0.070	-0.076
		qwen7b	-0.058	-0.029	0.022	0.052	0.054
	Ethnicity	black	llama70b	0.000	0.000	0.000	0.000
			olmo32b	-0.038	-0.048	-0.024	0.019
			qwen32b	0.021	0.000	0.000	0.000
			qwen7b	0.000	-0.000	-0.000	0.035
		hisp	llama70b	0.000	0.000	-0.000	-0.000
			olmo32b	0.131	-0.070	-0.105	0.058
			qwen32b	0.000	0.000	0.000	-0.000
			qwen7b	0.000	0.034	0.000	0.039
		middleEastern	llama70b	-0.000	-0.000	-0.000	0.000
			olmo32b	0.181	-0.031	0.062	-0.043
			qwen32b	0.083	0.000	0.000	0.000
			qwen7b	-0.000	-0.058	-0.000	-0.066
		other	llama70b	0.000	-0.000	0.000	0.000
			olmo32b	0.000	0.000	0.000	-0.000
			qwen32b	-0.000	0.000	0.000	—
			qwen7b	-0.000	-0.000	-0.000	0.000
		white	llama70b	—	0.000	0.000	0.000
			olmo32b	0.000	0.000	0.000	0.000
			qwen32b	0.000	-0.000	-0.000	0.000
			qwen7b	-0.000	-0.000	-0.000	0.000
Gender	man	llama70b	-0.027	-0.003	0.034	0.012	0.035
		olmo32b	-0.001	0.041	0.007	0.008	-0.037
		qwen32b	-0.058	0.148	0.000	-0.085	-0.034
		qwen7b	0.006	0.002	-0.006	-0.002	0.043
	nonBinary	llama70b	0.083	0.062	0.065	-0.000	-0.015

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Table 17. Aposteriori unimodality results for the LLM annotations of the Sap et al. 2022 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
		olmo32b	-0.028	-0.024	-0.025	-0.004	-0.002
		qwen32b	-0.043	-0.022	-0.049	0.016	0.035
		qwen7b	-0.003	0.023	-0.043	-0.009	-0.059
	woman	llama70b	-0.126	0.004	-0.093	0.048	0.006
		olmo32b	0.040	0.013	-0.009	0.028	0.012
		qwen32b	0.063	0.057	0.031	-0.000	0.000
		qwen7b	-0.013	0.003	0.033	-0.035	0.010

J The Ideal Subjective Dataset

We briefly discuss the type of data most suited for our proposed methodology, and connect it with current NLP datasets.

How Many Annotators? Our methodology does not directly require numerous annotators overall, only a sufficient number per item—indeed, statistically significant results can be obtained with as little as four annotations (see Sap-AA; Table 2). A large annotator pool however is required to get sufficient coverage on the examined groups. More annotators allow both higher fidelity of each PC dimension, and more such dimensions.

How Many Annotators per item? We investigate observed polarization against varying numbers of annotators. Specifically, for every item, we repeatedly sampled n annotators with replacement, calculated pol_{obs} for each sample and measured SD, averaging it across 1000 same-sized samples. As shown in Fig. 11, pol_{obs} exhibits a clear, inverse relationship with the number of annotators, attributed to better histogram estimation provided by nDFU (§5), though the rate of gain substantially diminishes after 20 annotators. Interestingly, some variability persists even with a very high annotator count.

How Many Items? Our methodology is designed to work with smaller datasets common in NLP. Three of the four datasets in the study feature low (300–600) item counts, while the Kumar dataset has been subsampled to 3% of its original size. Large datasets should be used only to provide larger coverage on the target domain, since a much better representation of individual groups can be achieved with more annotators per item, rather than more items. Smaller datasets may also be more practical for social science analysis [39].

Filtering Datasets with Desirable Levels of Subjectivity. In order to investigate whether the numerous annotators per items can explain the lack of UnPol in the DICES datasets (§4) we gradually subsample the annotators for each item randomly with replacement, starting from five annotators per item, and scaling up to the full complement of annotators in the dataset with a step of two. For each such step, we choose ten annotator samples and compute the maximum UnPol across all runs. Even with annotator counts similar to the Kumar and DICES datasets, UnPol remains zero, suggesting that the absence of UnPol can not be attributed to annotator number alone—rather, it is plausibly a pattern attributable to the DICES datasets themselves. This kind of analysis may be used in the exploratory phase of dataset selection in order to filter out datasets without enough group-specific subjectivity.

K Full Results

Tables 18,19,21,20 and 22 show the full apunim results for the Kumar, Sap, DICES-350, DICES-990 and POPQUORN datasets respectively. Note that the results for Kumar are unreliable since they rely on a small minority of items where

Table 18. Aposteriori unimodality results for the Kumar-results dataset. Stars indicate statistical significance: *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

PC	Value	apunim	support
Age	1) 18 - 24	0.0139	31
	2) 25 - 34	-0.0028	867
	3) 35 - 44	0.0000	279
	4) 45 - 54	0.0223	58
Education	2) High School graduate	0.0000	18
	3) College, no degree	0.0100	128
	4) Associate degree	0.0000	31
	5) Bachelor's degree	0.0013	947
Ethnicity	6) Master's degree	0.0000	68
	Black	0.0000	46
	White	0.0000	2622
Gender	Female	0.0010	1587
	Male	0.0006	1033
Has Been Targeted	False	0.0011	2540
	True	-0.0095	409
Is Parent	No	-0.0021	1089
	Yes	-0.0001	1603
Is Transgender	No	0.0000	4045
Political Affiliation	Conservative	-0.0134	260
	Independent	0.0000	349
	Liberal	-0.0010	850
Religion Important	1) No	0.0022	401
	2) Not very	0.0000	22
	3) Somewhat	0.0000	289
	4) Very	0.0002	463
Seen Toxicity	False	-0.0070	286
	True	-0.0008	2861
Sexual Orientation	Bisexual	-0.0678	50
	Heterosexual	-0.0010	3199
Technology Impact	2) Somewhat negative	0.0000	15
	3) Neutral	0.1243	27
	4) Somewhat positive	0.0038	1343
	5) Very positive	-0.0148	269
Toxicity Problem	1) Never	0.0000	7
	2) Rarely	0.0000	84
	3) Occasionally	0.0000	521
	4) Frequently	0.0000	422
	5) Very Frequently	0.0000	22

annotation counts happened to be over five, while our methodology requires six annotators per item to be reliable (see §5). We nevertheless provide the full results.

Table 19. Aposteriori unimodality results for the Sap-results dataset. Stars indicate statistical significance: *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

PC	Value	apunim	support
Age	1) Gen. X+	-0.1670*	223
	2) Millennial	0.0199	1651
Ethnicity	white	-0.0007	1235
Gender	man	-0.1508***	1254
	woman	0.2613***	693

Table 20. Aposteriori unimodality results for the Dices-350-results dataset. Stars indicate statistical significance: *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

PC	Value	apunim	support
Age	1) Gen. Z	-0.0043	14728
	2) Millennial	-0.0089	9468
	3) Gen. X+	0.0148	8153
Education	College +	0.0105	19725
	High school -	-0.0145	10783
	Other	-0.0230*	1841
Gender	Man	-0.0171	16043
	Woman	0.0176	16306
Race	African Am.	-0.0366***	7627
	Asian	0.0445***	6838
	Latino	0.0059	5786
	Multiracial	-0.0061	4208
	White	-0.0113	7890

L Prompts Used

This section only involves the full contents of the prompts. See App. I for a high-level description of the prompt variants and what each aims to accomplish.

L.1 Main Annotation Prompts

L.1.1 Kumar.

Default. You are an annotator with the following characteristics: {{persona}} How toxic do you personally find this comment? 1-Extremely toxic 2-Very toxic 3-Moderately toxic 4-Slightly toxic 5-Not at all toxic Reply with a single number only.

Persona-focused. You are an annotator with the following characteristics: persona
Annotate the comment according to how your persona would personally perceive it. Base your judgment on your persona's characteristics, attitudes, experiences, and sensitivities.
How toxic would your persona find this comment?

Table 21. Aposteriori unimodality results for the Dices-990-results dataset. Stars indicate statistical significance: *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

PC	Value	apunim	support
Age	1) Gen. Z	0.0469***	12214
	2) Millennial	-0.0068*	34433
	3) Gen. X+	-0.0282***	12699
Education	College +	0.0063*	52727
	High school -	-0.0500***	6619
Gender	Man	0.0359***	28817
	Woman	-0.0271***	29990
Race	African Am.	0.0995***	5148
	Asian	-0.0057	16486
	Latino	-0.0631***	5877
	Other	0.0340***	21669
	White	-0.0649***	10166

(1) Extremely toxic

(2) Very toxic

(3) Moderately toxic

(4) Slightly toxic

(5) Not at all toxic

Reply with a single number only.

Adversarial. You are an annotator with the following characteristics: persona

When rating the comment, be very sensitive to the specific characteristics of your persona. Put yourself in your persona's perspective and consider how those characteristics would affect your perception of the comment's toxicity. Give substantial weight to your persona's particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person's judgment.

How toxic would your persona find this comment?

(1) Extremely toxic

(2) Very toxic

(3) Moderately toxic

(4) Slightly toxic

(5) Not at all toxic

Reply with a single number only.

Direct. You must process this comment using only the characteristics defined in persona. The persona's values are the primary, mandatory filter for this task. Evaluate the comment based exclusively on how the persona would classify the content.

How toxic would your persona find this comment?

(1) Extremely toxic

Table 22. Aposteriori unimodality results for the Popquorn-results dataset. Stars indicate statistical significance: *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

PC	Value	apunim	support
Age	01) 18-24	-0.1198	90
	02) 25-29	0.0420	50
	03) 30-34	0.2251	71
	04) 35-39	0.0439	121
	05) 40-44	0.0061	81
	06) 45-49	0.0507	64
	07) 50-54	0.0381	22
	08) 54-59	-0.0635	133
	09) 60-64	0.1194	13
	10) >65	-0.0145	118
Education	2) High school	-0.0112	1270
	3) College degree	0.0496***	2388
	4) Graduate degree	-0.0808**	468
Gender	Man	0.0081	2720
	Woman	-0.0195	2899
Occupation	Employed	-0.0126	2535
	Homemaker	-0.2366	12
	Retired	-0.1026	95
	Self-employed	0.1063*	210
	Student	0.3675	18
	Unemployed	0.1204*	166
Race	Asian	0.2795*	29
	Black	-0.1572***	222
	White	0.0090	4135

- (2) Very toxic
- (3) Moderately toxic
- (4) Slightly toxic
- (5) Not at all toxic

Reply with a single number only.

Single. Your assigned characteristic set is persona. Identify the single, most influential trait from your persona (e.g., 'Intolerance for X,' 'Sensitivity to Y,' 'Strong belief in Z').

Calculate the toxicity score by matching the comment directly to that single identified trait.

Based on this sole variable, what is the toxicity score?

- (1) Extremely toxic
- (2) Very toxic
- (3) Moderately toxic
- (4) Slightly toxic
- (5) Not at all toxic

L.1.2 Sap.

Default. You are an annotator with the following characteristics: {{persona}} In your opinion, how racist is this post? Use a scale ranging from “not at all” (1) to “very much so” (5). Reply with a single number only.

Persona-focused. You are an annotator with the following characteristics: persona

Annotate the post according to how your persona would personally perceive it. Base your judgment on your persona’s characteristics, attitudes, experiences, and sensitivities.

In your opinion, how racist would your persona find this post? Use a scale ranging from “not at all” (1) to “very much so” (5).

Reply with a single number only.

Adversarial. You are an annotator with the following characteristics: persona

When rating the post, be very sensitive to the specific characteristics of your persona. Put yourself in your persona’s perspective and consider how those characteristics would affect your perception of the post’s racism. Give substantial weight to your persona’s particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person’s judgment.

In your opinion, how racist would your persona find this post? Use a scale ranging from “not at all” (1) to “very much so” (5).

Reply with a single number only.

Direct. You must process this comment using only the characteristics defined in persona. The persona’s values are the primary, mandatory filter for this task. Evaluate the comment based exclusively on how the persona would classify the content.

In your opinion, how racist would your persona find this post? Use a scale ranging from “not at all” (1) to “very much so” (5).

Reply with a single number only.

Single. Your assigned characteristic set is persona. Identify the single, most influential trait from your persona (e.g., ‘Intolerance for X,’ ‘Sensitivity to Y,’ ‘Strong belief in Z’).

Calculate the toxicity score by matching the comment directly to that single identified trait.

Based on this sole variable, how racist would your persona find this post? Use a scale ranging from “not at all” (1) to “very much so” (5).

Reply with a single number only.

L.2 Ablation Annotation Prompts

L.2.1 Kumar.

Variant 1. You are an annotator with the following characteristics: {{persona}}

How toxic do you personally find this comment? 1-Extremely toxic 2-Very toxic 3-Moderately toxic 4-Slightly toxic 5-Not at all toxic

Reply with a single number only.

Variant 2. You are an annotator with the following characteristics: {{persona}}.

Evaluate how toxic this comment is from your personal perspective. Use exactly one of the following ratings:

- 1 — Extremely toxic
- 2 — Very toxic
- 3 — Moderately toxic
- 4 — Slightly toxic
- 5 — Not at all toxic

Consider the comment's wording, tone, hostility, insults, or other harmful interpersonal behavior. Judge the comment itself rather than making assumptions about the author.

Return only a single number.

Variant 3. You are an annotator with the following characteristics: {{persona}}.

Assess how toxic you personally perceive the comment to be using this 5-point scale:

- 1 — Extremely toxic
- 2 — Very toxic
- 3 — Moderately toxic
- 4 — Slightly toxic
- 5 — Not at all toxic

Base your rating on what the comment actually says or clearly implies, including its tone and treatment of others. Do not infer toxicity solely from the topic under discussion.

Respond with only the corresponding number: 1, 2, 3, 4, or 5.

L.2.2 Sap.

Variant 1. You are an annotator with the following characteristics: {persona}.

Evaluate the post and rate how racist it is on a 5-point scale:

- 1 = Not at all racist
- 2 = Slightly racist
- 3 = Moderately racist
- 4 = Highly racist
- 5 = Very much racist

Base your rating on the content of the post itself. Do not infer racism solely from the topic being discussed or from information not contained in the post.

Reply with a single number only.

Variant 2. You are an annotator with the following characteristics: {{persona}}.

In your judgment, how strongly does this post express or promote racism? Assign one rating from 1 to 5:

- 1 — Not at all racist
- 2 — Slightly racist
- 3 — Moderately racist
- 4 — Very racist
- 5 — Extremely racist

Consider only what the post says or clearly implies. Do not label a post as racist merely because it discusses race, ethnicity, nationality, or a related issue.

Respond with one number and nothing else.

Variant 3. You are an annotator with the following characteristics: {{persona}}.

Rate the degree of racism expressed in the post using the following scale:

- 1 – Not at all
- 2 – A little
- 3 – Somewhat
- 4 – Quite a lot
- 5 – Very much so

Use the wording and meaning of the post as the basis for your assessment. Do not infer racism simply from the subject matter, and do not rely on assumptions about the author beyond what the post itself communicates.

Output only the selected number: 1, 2, 3, 4, or 5.

References

- [1] Gavin Abercrombie, Tanvi Dinkar, Amanda Cercas Curry, Verena Rieser, and Dirk Hovy. 2025. Consistency is Key: Disentangling Label Variation in Natural Language Processing with Intra-Annotator Agreement. In *Proceedings of the 4th Workshop on Perspectivist Approaches to NLP*, Gavin Abercrombie, Valerio Basile, Simona Frenda, Sara Tonelli, and Shiran Dudy (Eds.). Association for Computational Linguistics, Suzhou, China, 63–74. doi:10.18653/v1/2025.nlperspectives-1.6
- [2] Sohail Akhtar, Valerio Basile, and Viviana Patti. 2019. A New Measure of Polarization in the Annotation of Hate Speech. In *AI*IA 2019 – Advances in Artificial Intelligence*, Mario Alviano, Gianluigi Greco, and Francesco Scarcello (Eds.). Springer International Publishing, Cham, 588–603.
- [3] Lora Aroyo, Alex Taylor, Mark Diaz, Christopher Homan, Alicia Parrish, Gregory Serapio-Garcia, Vinodkumar Prabhakaran, and Ding Wang. 2023. DICES Dataset: Diversity in Conversational AI Evaluation for Safety. In *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (Eds.), Vol. 36. Curran Associates, Inc., 53330–53342. https://proceedings.neurips.cc/paper_files/paper/2023/file/a74b697bce4cac6c91896372abaa8863-Paper-Datasets_and_Benchmarks.pdf
- [4] Reuben Binns, Michael Veale, Max Van Kleek, and Nigel Shadbolt. 2017. Like trainer, like bot? Inheritance of bias in algorithmic content moderation. In *International conference on social informatics*. Springer, 405–415.
- [5] James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. 2024. Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. *Political Analysis* 32, 4 (2024), 401–416. doi:10.1017/pan.2024.5
- [6] Ted Byrt, Janet Bishop, and John B. Carlin. 1993. Bias, prevalence and kappa. *Journal of Clinical Epidemiology* 46, 5 (01 May 1993), 423–429. doi:10.1016/0895-4356(93)90018-V
- [7] Alessandro Checco, A. Roitero, Eddy Maddalena, S. Mizzaro, and Gianluca Demartini. 2017. Let’s Agree to Disagree: Fixing Agreement Measures for Crowdsourcing.
- [8] S.Y. Chen, Z. Feng, and X. Yi. 2017. A general introduction to adjustment for multiple comparisons. *Journal of Thoracic Disease* 9, 6 (2017), 1725–1729. doi: 10.21037/jtd.2017.05.34; PMID: 28740688; PMCID: PMC5506159.
- [9] Aida Mostafazadeh Davani, Mohammad Atari, Brendan Kennedy, and Morteza Dehghani. 2023. Hate Speech Classifiers Learn Normative Social Stereotypes. *Transactions of the Association for Computational Linguistics* 11 (2023), 300–319. doi:10.1162/tacl_a_00550
- [10] Yijiang River Dong, Tiancheng Hu, and Nigel Collier. 2024. Can LLM be a Personalized Judge?. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 10126–10141. doi:10.18653/v1/2024.findings-emnlp.592
- [11] European Union. 2016. Regulation (EU) 2016/679 of the European Parliament and of the Council. <https://data.europa.eu/eli/reg/2016/679/oj>
- [12] European Union. 2024. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>. <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng> Official Journal of the European Union, L 2024/1689.
- [13] Alvan R. Feinstein and Domenic V. Cicchetti. 1990. High agreement but low Kappa: I. the problems of two paradoxes. *Journal of Clinical Epidemiology* 43, 6 (01 Jan 1990), 543–549. doi:10.1016/0895-4356(90)90158-L
- [14] Eve Fleisig, Rediet Abebe, and Dan Klein. 2023. When the Majority is Wrong: Modeling Annotator Disagreement for Subjective Tasks. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 6715–6726. doi:10.18653/v1/2023.emnlp-main.415

Manuscript submitted to ACM

- [15] Juan Felipe Gomez, Caio Machado, Lucas Monteiro Paes, and Flavio Calmon. 2024. Algorithmic Arbitrariness in Content Moderation. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency* (Rio de Janeiro, Brazil) (FAccT '24). Association for Computing Machinery, New York, NY, USA, 2234–2253. doi:10.1145/3630106.3659036
- [16] Nitesh Goyal, Ian D Kivlichan, Rachel Rosen, and Lucy Vasserman. 2022. Is your toxicity my toxicity? exploring the impact of rater identity on toxicity annotation. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (2022), 1–28.
- [17] Aaron Grattafiori et al. 2024. The Llama 3 Herd of Models. arXiv:2407.21783 [cs.AI] <https://arxiv.org/abs/2407.21783>
- [18] Dirk Groeneveld et al. 2024. OLMo: Accelerating the Science of Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Bangkok, Thailand, 15789–15809. doi:10.18653/v1/2024.acl-long.841
- [19] Soumyajit Gupta, Sooyong Lee, Maria De-Arteaga, and Matthew Lease. 2023. Same Same, But Different: Conditional Multi-Task Learning for Demographic-Specific Toxicity Detection. In *Proceedings of the ACM Web Conference 2023* (Austin, TX, USA) (WWW '23). Association for Computing Machinery, New York, NY, USA, 3689–3700. doi:10.1145/3543507.3583290
- [20] Shashank Gupta, Vaishnavi Shrivastava, Ameet Deshpande, Ashwin Kalyan, Peter Clark, Ashish Sabharwal, and Tushar Khot. 2024. Bias runs deep: Implicit reasoning biases in persona-assigned llms. In *International Conference on Learning Representations*, Vol. 2024. 21849–21874.
- [21] Sture Holm. 1979. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics* 6, 2 (1979), 65–70. <http://www.jstor.org/stable/4615733>
- [22] Jonathan Ivey, Susan Gauch, and David Jurgens. 2025. NUTMEG: Separating Signal From Noise in Annotator Disagreement. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (Eds.). Association for Computational Linguistics, Suzhou, China, 2874–2887. doi:10.18653/v1/2025.emnlp-main.144
- [23] Shivani Kapania, Stephanie Ballard, Alex Kessler, and Jennifer Wortman Vaughan. 2025. Examining the Expanding Role of Synthetic Data Throughout the AI Development Pipeline. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT '25)*. Association for Computing Machinery, New York, NY, USA, 45–60. doi:10.1145/3715275.3732005
- [24] Petra Kralj Novak, Teresa Scantamburlo, Andraž Pelicon, Matteo Cinelli, Igor Mozetič, and Fabiana Zollo. 2022. Handling Disagreement in Hate Speech Modelling. In *Information Processing and Management of Uncertainty in Knowledge-Based Systems*, Davide Ciucci, Inés Couso, Jesús Medina, Dominik Ślęzak, Davide Petturiti, Bernadette Bouchon-Meunier, and Ronald R. Yager (Eds.). Springer International Publishing, Cham, 681–695.
- [25] Deepak Kumar, Patrick Gage Kelley, Sunny Consolvo, Joshua Mason, Elie Bursztein, Zakir Durumeric, Kurt Thomas, and Michael Bailey. 2021. Designing toxic content classification for a diversity of perspectives. In *Proceedings of the Seventeenth USENIX Conference on Usable Privacy and Security (SOUPS'21)*. USENIX Association, USA, Article 16, 19 pages.
- [26] Noah Lee, Na Min An, and James Thorne. 2023. Can Large Language Models Capture Dissenting Human Voices?. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 4569–4585. doi:10.18653/v1/2023.emnlp-main.278
- [27] Elisa Leonardelli, Gavin Abercrombie, Dina Almanea, Valerio Basile, Tommaso Fornaciari, Barbara Plank, Verena Rieser, Alexandra Uma, and Massimo Poesio. 2023. SemEval-2023 Task 11: Learning with Disagreements (LeWiDi). In *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*, Atul Kr. Ojha, A. Seza Doğruöz, Giovanni Da San Martino, Harish Tayyar Madabushi, Ritesh Kumar, and Elisa Sartori (Eds.). Association for Computational Linguistics, Toronto, Canada, 2304–2318. doi:10.18653/v1/2023.semeval-1.314
- [28] Andy Liu, Mona Diab, and Daniel Fried. 2024. Evaluating Large Language Model Biases in Persona-Steered Generation. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 9832–9850. doi:10.18653/v1/2024.findings-acl.586
- [29] Josephine Lukito, Bin Chen, Gina M. Masullo, and Natalie Jomini Stroud. 2024. Comparing a BERT Classifier and a GPT classifier for Detecting Connective Language Across Multiple Social Media. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 19140–19153. doi:10.18653/v1/2024.emnlp-main.1067
- [30] Marlene Lutz, Indira Sen, Georg Ahnert, Elisa Rogers, and Markus Strohmaier. 2025. The Prompt Makes the Person(a): A Systematic Evaluation of Sociodemographic Persona Prompting for Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (Eds.). Association for Computational Linguistics, Suzhou, China, 23212–23237. doi:10.18653/v1/2025.findings-emnlp.1261
- [31] Sarah Masud, Sahajpreet Singh, Viktor Hangya, Alexander Fraser, and Tanmoy Chakraborty. 2024. Hate Personified: Investigating the role of LLMs in content moderation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 15847–15863. doi:10.18653/v1/2024.emnlp-main.886
- [32] Giuseppe Mignemi, Antonio Calcagni, Andrea Spoto, and Ioanna Manolopoulou. 2024. Mixture polarization in inter-rater agreement analysis: a Bayesian nonparametric index. *Statistical Methods & Applications* 33, 1 (01 Mar 2024), 325–355. doi:10.1007/s10260-023-00741-x
- [33] Aida Mostafazadeh Davani, Mark Díaz, and Vinodkumar Prabhakaran. 2022. Dealing with Disagreements: Looking Beyond the Majority Vote in Subjective Annotations. *Transactions of the Association for Computational Linguistics* 10 (2022), 92–110. doi:10.1162/tacl_a_00449
- [34] Team Olmo. 2026. Olmo 3. arXiv:2512.13961 [cs.CL] <https://arxiv.org/abs/2512.13961>
- [35] John Pavlopoulos and Aristidis Likas. 2024. Polarized Opinion Detection Improves the Detection of Toxic Language. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, Yvette Graham and Matthew Purver

- (Eds.). Association for Computational Linguistics, St. Julian's, Malta, 1946–1958. <https://aclanthology.org/2024.eacl-long.117/>
- [36] Jiaxin Pei and David Jurgens. 2023. When Do Annotator Demographics Matter? Measuring the Influence of Annotator Demographics with the POPQUORN Dataset. In *Proceedings of the 17th Linguistic Annotation Workshop (LAW-XVII)*, Jakob Prange and Annemarie Friedrich (Eds.). Association for Computational Linguistics, Toronto, Canada, 252–265. doi:10.18653/v1/2023.law-1.25
- [37] Anne Quaranto. 2022. Dog whistles, covertly coded speech, and the practices that enable them. *Synthese* 200, 4 (2022), 330.
- [38] Amirhossein Razavi, Mina Soltangheis, Negar Arabzadeh, Sara Salamat, Morteza Zihayat, and Ebrahim Bagheri. 2025. Benchmarking Prompt Sensitivity in Large Language Models. In *Advances in Information Retrieval*, Claudia Hauff, Craig Macdonald, Dietmar Jannach, Gabriella Kazai, Franco Maria Nardini, Fabio Pinelli, Fabrizio Silvestri, and Nicola Tonello (Eds.). Springer Nature Switzerland, Cham, 303–313.
- [39] Luca Rossi, Katherine Harrison, and Irina Shklovski. 2024. The Problems of LLM-generated Data in Social Science Research. *Sociologica* 18, 2 (1 2024), 145–168. doi:10.6092/issn.1971-8853/19576
- [40] Paul Röttger, Bertie Vidgen, Dirk Hovy, and Janet Pierrehumbert. 2022. Two Contrasting Data Annotation Paradigms for Subjective NLP Tasks. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Marine Carpuat, Marie-Catherine de Marneffe, and Ivan Vladimir Meza Ruiz (Eds.). Association for Computational Linguistics, Seattle, United States, 175–190. doi:10.18653/v1/2022.naacl-main.13
- [41] Maarten Sap, Dallas Card, Saadia Gabriel, Yejin Choi, and Noah A. Smith. 2019. The Risk of Racial Bias in Hate Speech Detection. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Anna Korhonen, David Traum, and Lluís Màrquez (Eds.). Association for Computational Linguistics, Florence, Italy, 1668–1678. doi:10.18653/v1/P19-1163
- [42] Maarten Sap, Swabha Swayamdipta, Laura Vianna, Xuhui Zhou, Yejin Choi, and Noah A. Smith. 2022. Annotators with Attitudes: How Annotator Beliefs And Identities Bias Toxic Language Detection. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Marine Carpuat, Marie-Catherine de Marneffe, and Ivan Vladimir Meza Ruiz (Eds.). Association for Computational Linguistics, Seattle, United States, 5884–5906. doi:10.18653/v1/2022.naacl-main.431
- [43] Ole Tange. 2025. GNU Parallel 20250822. doi:10.5281/zenodo.16944306 GNU Parallel is a general parallelizer to run multiple serial command line programs in parallel without changing them..
- [44] Alexandra Uma, Dina Almanea, and Massimo Poesio. 2022. Scaling and Disagreements: Bias, Noise, and Ambiguity. *Frontiers in Artificial Intelligence* Volume 5 - 2022 (2022). doi:10.3389/frai.2022.818451
- [45] Alexandra Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2021. Learning from Disagreement: A Survey. *Journal of Artificial Intelligence Research* 72 (12 2021), 1385–1470. doi:10.1613/jair.1.12752
- [46] Mariken A. C. G. van der Velden, Felicia Loecherbach, Wouter van Atteveldt, Antske Fokkens, Myrthe Reuver, and Kasper Welbers. 2025. Whose truth is it anyway? An experiment on annotation bias in times of factual opinion polarization. *Communication Methods and Measures* 19, 4 (2025), 332–349. arXiv:<https://doi.org/10.1080/19312458.2025.2562034> doi:10.1080/19312458.2025.2562034
- [47] Nikolas Vitsakis, Amit Parekh, Tanvi Dinkar, Gavin Abercrombie, Ioannis Konstas, and Verena Rieser. 2023. iLab at SemEval-2023 Task 11 Le-Wi-Di: Modelling Disagreement or Modelling Perspectives?. In *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*, Atul Kr. Ojha, A. Seza Doğruöz, Giovanni Da San Martino, Harish Tayyar Madabushi, Ritesh Kumar, and Elisa Sartori (Eds.). Association for Computational Linguistics, Toronto, Canada, 1660–1669. doi:10.18653/v1/2023.semeval-1.231
- [48] Nikolas Vitsakis, Amit Parekh, and Ioannis Konstas. 2024. Voices in a Crowd: Searching for clusters of unique perspectives. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 12517–12539. doi:10.18653/v1/2024.emnlp-main.696
- [49] Franziska Weeber, Vera Neplenbroek, Jan Batzner, and Sebastian Padó. 2026. One Persona, Many Cues, Different Results: How Sociodemographic Cues Impact LLM Personalization. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Maria Liakata, Viviane P. Moreira, Jiajun Zhang, and David Jurgens (Eds.). Association for Computational Linguistics, San Diego, California, United States, 44892–44921. doi:10.18653/v1/2026.acl-long.2079
- [50] Michael Wojatzki, Tobias Horsmann, Darina Gold, and Torsten Zesch. 2018. Do women perceive hate differently: Examining the relationship between hate speech, gender, and agreement judgments. (2018).
- [51] Ka Wong, Praveen Paritosh, and Lora Aroyo. 2021. Cross-replication Reliability - An Empirical Approach to Interpreting Inter-rater Reliability. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli (Eds.). Association for Computational Linguistics, Online, 7053–7065. doi:10.18653/v1/2021.acl-long.548
- [52] An Yang et al. 2025. Qwen2.5 Technical Report. arXiv:2412.15115 [cs.CL] <https://arxiv.org/abs/2412.15115>
- [53] Caleb Ziems, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang. 2024. Can large language models transform computational social science? *Computational linguistics* 50, 1 (2024), 237–291.

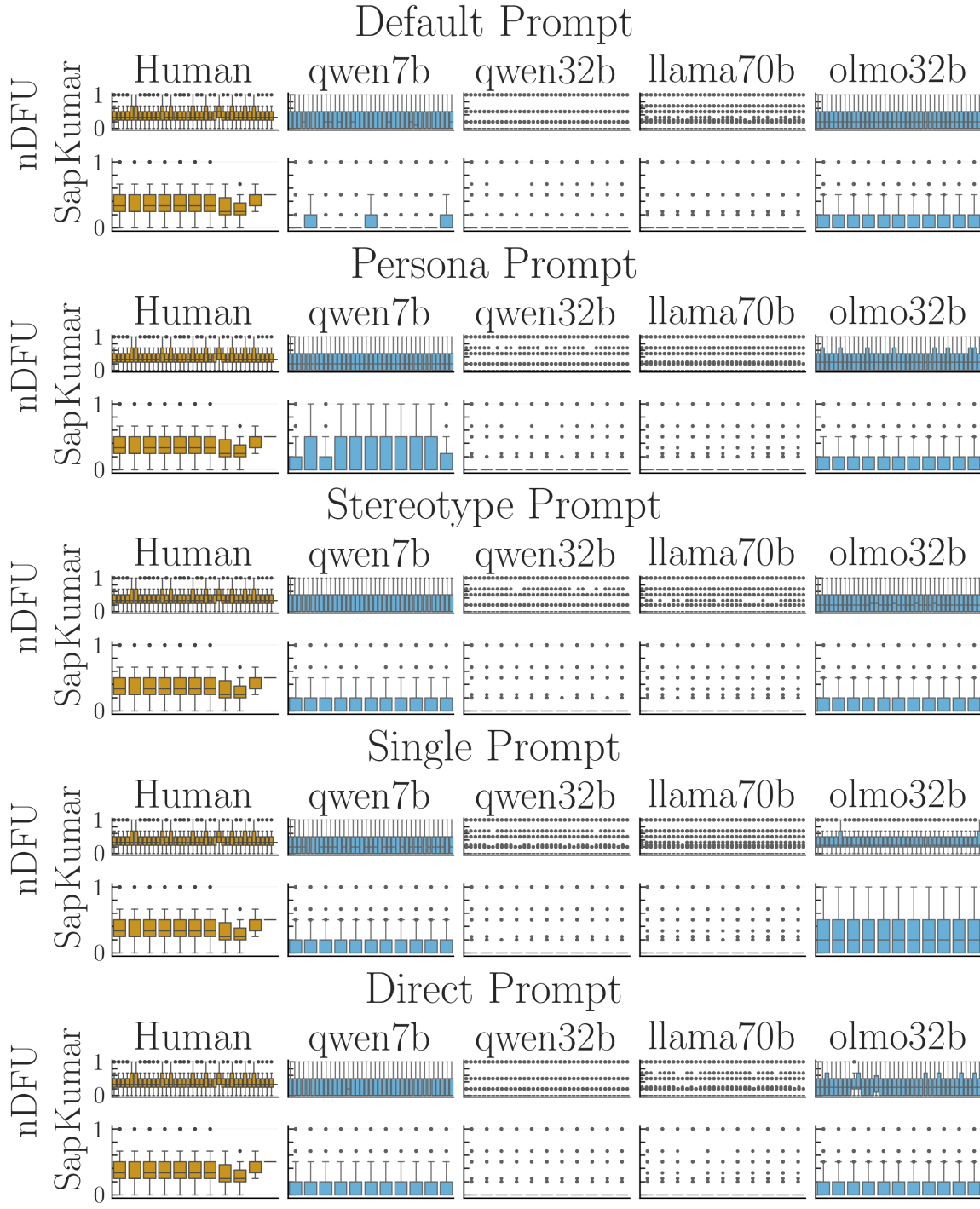


Fig. 10. nDFU grouped by dataset (rows) and model family (columns) for each PC persona dimension in the primary annotation task (blue) using different instruction prompts, compared with human annotations on the same sample (yellow).

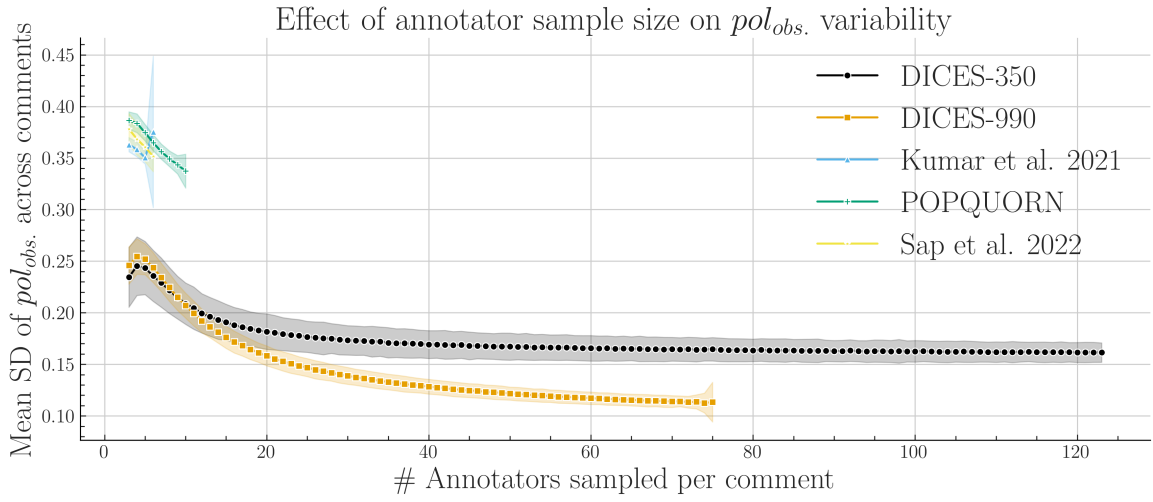


Fig. 11. SD of pol_{obs} across items, averaged across 1000 runs. Shaded areas represent two SDs. Pol. estimates benefit from more annotators up to $n = 20$, at which point they begin to plateau.